AI500: Foundations of Artificial Intelligence

Module 1 Quiz & Assignment Preparation Guide

What is expected, what is likely to come, and full answers to every end-of-module question

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Sources: HMD Foundations of AI Notes 2026 (Module 1), Assignment 1 Report, PA1 Notebook

Answers follow the wording, examples and templates of the course notes wherever possible.

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Part 1: How to Use This Guide

This guide is built from three sources: the course notes (HMD Foundations of AI Notes 2026, Module 1), your Assignment 1 report and your PA1 notebook. Wherever possible, the definitions, lists, examples and templates are taken from the notes, so that the wording in your answers matches what sir taught.

What the quiz will expect from you

The practice questions at the end of the notes fall into four styles. The quiz is very likely to use the same styles:

- **Short conceptual questions (Q1 to Q100):** define a term, compare two ideas, or explain why something matters. Answer with the definition from the notes, one reason, and one example from the notes.
- **Scenario questions (S1 to S5):** a short case is given and you must diagnose it: find the leakage, the train-serving mismatch, the feedback loop, the missing framing elements, or the right model. Use the exact vocabulary (index time, prediction horizon, temporal leakage, selective labels, calibration, subgroup performance and so on).
- **Reflection questions (R1 to R5):** long answers that apply the whole pipeline to one system. These are good practice for longer quiz questions and assignment write-ups.
- **Application problems (A1 to A7):** produce an artifact: a problem formulation, a **Model Card**, a **Datasheet**, a model-selection decision, a **technical-debt register**, a **problem-framing worksheet** or a **failure analysis**. Your assignment contained four of these (worksheet, model card, datasheet, failure analysis), so expect one of them in the quiz.
- **One question from your report or notebook:** Part 3 of this guide recaps the assignment and gives worked answers to the questions most likely to be asked about it (including hand calculations for the MLP forward pass and Bayes' theorem).

How to write a high-scoring answer

- **Start with the decision, not the algorithm.** Almost every answer in this module improves when you say who decides, when, and what action follows.
- **Name the concept, then apply it.** For example: "This is temporal leakage, because the antibiotic count is recorded after the sepsis diagnosis, so it is not available at the index time."
- **Always mention the four "beyond accuracy" checks:** calibration, subgroup performance, comparison with a baseline, and decision utility.
- **Always mention the human side:** human review, abstention, appeal, monitoring, and a named accountable owner (NIST Govern).
- **Give a recommendation when asked.** Model cards end with approve, conditionally approve, or do not approve. Failure analyses end with proceed, redesign, replace or cancel.

> **Exam tip:** Sir said the end-of-module questions will help in both the quiz and the assignment. Parts 4 to 8 answer every one of them (Q1 to Q100, S1 to S5, R1 to R5, A1 to A7). Part 2 is the fast revision sheet: if you only have one hour, read Part 2, Part 3 and Part 9.

Part 2: Must-Know Frameworks and Lists

This part collects every list, framework and formula from the notes that a question could ask you to reproduce. Learn these by heart.

1. Intellectual roots of AI (Section 1.1)

The history of AI is "the history of successive attempts to formalize intelligence", not simply the history of faster computers. The traditions that fed into AI:

- formal logic and the mechanization of reasoning;
- probability and statistical inference;
- computation and algorithms;
- information theory;
- control and feedback;
- cognitive science and theories of intelligence;
- operations research and optimization;
- philosophy of mind, language, knowledge, and agency.

The deeper question: "Can perception, reasoning, learning, decision-making, and intelligent action be represented sufficiently precisely to be carried out by a machine?"

> **From the notes:** Intelligence requires representation. A machine cannot reason directly about an unrepresented world. It reasons over variables, symbols, vectors, tokens, states, graphs, probabilities, or other constructed representations.

2. Key people and their contributions

Person / event	Contribution in the notes
Norbert Wiener (1948)	Cybernetics: connected communication, feedback and control in animals and machines. The consequences of previous actions are observed and used to modify future behaviour.
Claude Shannon (1948)	Information theory: formalized information, uncertainty, coding and transmission. Gave ML entropy, cross-entropy, mutual information, compression, channel capacity and noisy communication. Lesson: information is not automatically meaning.
Alan Turing (1950)	Reframed "Can machines think?" into an operational test based on observable behaviour (the Turing Test). Behavioural success does not settle whether a machine understands.
Dartmouth proposal (1955, workshop 1956)	McCarthy, Minsky, Rochester and Shannon. Used the term "artificial intelligence"; consolidated, named and legitimized symbolic reasoning, search and planning, ML, NLP, vision, robotics, knowledge representation and theorem proving.
Newell and Simon (1976)	Physical symbol systems are central to intelligent action (symbolic AI).
Judea Pearl	Causal framework: causal claims need explicit structural assumptions and mechanistic models; they cannot be derived from observational associations alone.
Russell and Norvig	Four ways of thinking about AI: think like humans, act like humans, think rationally, act rationally.
Mitchell, Gebru et al. (2019)	Model Cards for Model Reporting.
Gebru et al. (2018)	Datasheets for Datasets.

Person / event	Contribution in the notes
Sculley et al. (2015)	Hidden Technical Debt in Machine Learning Systems.
Obermeyer et al. (2019)	Health-care algorithm used cost as a proxy for need, which disadvantaged Black patients.

3. Symbolic vs statistical AI

Symbolic AI strengths	Symbolic AI limitations
explicit representations inspectable reasoning chains incorporation of domain rules compositional structures strong performance when the state space and rules are well specified	difficulty acquiring and maintaining knowledge brittleness under ambiguity poor handling of noisy perception combinatorial explosion dependence on manually encoded representations
Statistical ML strengths	**Statistical ML limitations**
learning from high-dimensional observations tolerance of noisy inputs adaptation from examples strong performance in perception, prediction and generation	dependence on data quality limited causal interpretation potential opacity vulnerability to distribution shift difficulty guaranteeing behaviour outside the observed data regime

Modern systems are often **hybrid**: learned perception with rule-based constraints; neural representations with search; language models with retrieval; statistical prediction with optimization; learned policies with safety controllers; machine-generated recommendations with human review. "Symbolic versus statistical should not be treated as a permanent binary."

4. Feedback risks (predictive policing example)

- **Self-reinforcing bias:** existing patterns in historical data become amplified over time.
- **Selective labels:** the system only observes outcomes where attention was directed, not the full reality.
- **Performative effects:** a prediction influences behaviour and creates conditions that make the prediction appear accurate.
- **Feedback loops:** the output of one prediction becomes part of the input for future predictions.

For ranking and recommendation: **Past exposure → Observed interaction → Training data → Future exposure.**

5. Four central philosophical questions (Section 1.2)

- **What counts as intelligence?** A spectrum of capabilities (reason, learn, solve unfamiliar problems, use language, plan, act toward goals, adapt, understand context, interact socially, create), not one trait.
- **Is prediction the same as understanding?** No. Prediction answers "What is likely to happen?"; causal reasoning answers "What would happen if a specific intervention were applied?"
- **Can intelligence be separated from embodiment and society?** No. A deployed system is a **socio-technical composite** of five layers: **Model, Data Infrastructure, Interface, Human Workflow, Institutional Rules.**
- **Who is responsible?** Not the algorithm. Responsibility belongs to people across the lifecycle; governance through the **NIST AI RMF**.

6. NIST AI Risk Management Framework

Function	Meaning (notes wording)
Govern	Establishing the organizational culture, accountability structures, and policies necessary to manage AI risk.
Map	Identifying, contextualizing, and understanding the specific risks associated with the AI system and its operational environment.
Measure	Assessing, analyzing, and tracking the identified risks using both quantitative and qualitative metrics.
Manage	Prioritizing and acting upon the risks, allocating resources to mitigate, transfer, or accept them.

Trustworthiness characteristics: valid and reliable; safe; secure and resilient; accountable and transparent; explainable and interpretable; privacy-enhanced; fair with harmful bias managed.

7. Definition of AI and the four classical views

> **From the notes:** AI is "the scientific and engineering discipline concerned with constructing computational systems that perceive or receive information, form useful representations, reason or learn from evidence, and select actions or outputs in pursuit of specified objectives under uncertainty and constraints."

View (Russell and Norvig)	Key idea
Think like humans	Reproduce human cognitive processes (roots: cognitive psychology, neuroscience, linguistics, philosophy of mind, computational modeling). Judged on whether it reaches answers through human-like mechanisms.
Act like humans	Behavioural: the Turing Test. Can behaviour be indistinguishable from a human's?
Think rationally	Formalize correct reasoning through logic (theorem proving, expert systems, logical inference). Limited because real environments are rarely complete or deterministic.
Act rationally	A rational agent selects actions expected to advance its objectives given its information and constraints. "Rational action does not mean perfect action."

8. Spectrum of autonomy

Level	Meaning	Example from notes
Passive analytical	Transforms data into insights or probabilities; cannot act.	Sales dashboard; disease-probability model; equipment-failure predictor.
Decision-support	Proposes actions; final authority stays with humans.	Specialist-referral recommender; loan ranking for underwriter review.
Semi-autonomous	Acts automatically in routine low-risk cases; escalates ambiguous or high-stakes cases.	Fraud system that blocks clear fraud and routes borderline cases to investigators.
Autonomous	Perceives, decides and acts with little or no human intervention.	Manufacturing robots; warehouse sorting; high-frequency trading.

"Autonomy must not be conflated with intelligence": a thermostat is autonomous but not intelligent; an LLM may be linguistically intelligent but has no agency.

9. The AI ecosystem (Table 1)

Field	Central question	Relationship to AI
Artificial Intelligence	How can machines perform tasks involving perception, reasoning, learning, planning, language, decision-making or adaptive action?	Broad umbrella
Machine Learning	How can a system improve a task-related function by learning patterns or parameters from data or experience?	Major approach within AI
Data Science	How can data be acquired, prepared, analyzed, modeled, interpreted and communicated to produce knowledge or decisions?	Broader lifecycle that may or may not use AI
Statistics	What can be inferred about a population or process from uncertain, variable or incomplete observations?	Mathematical and inferential foundation
Analytics	What happened, why, what may happen, and what action should be considered?	Decision-oriented use of data
Optimization	Which feasible choice minimizes or maximizes an objective?	Used in training, planning, allocation, control
Automation	How can a predefined process be executed with limited human intervention?	May contain no learning or intelligence
Generative AI	How can a model produce new content conditioned on data, prompts or context?	A family of AI capabilities, not a synonym for all AI
Data Engineering	How can data be collected, represented, validated, transformed, governed, delivered and monitored reliably at scale?	Infrastructure on which data-driven AI depends

NIST Generative AI risks: confabulation (hallucinations); harmful content; information integrity (deepfakes); intellectual property and privacy; misuse.

Optimization is ethically and epistemically blind. It cannot tell whether y is the correct target, whether the sample represents the deployment population, whether the loss reflects real harm, whether the model should be deployed, or whether the decision is ethically acceptable. "Optimization solves the objective that was specified. It does not validate the objective."

$$\hat{R}(\theta) = (1/n)\ \Sigma\ L(\ f_\theta(x_i), y_i\ )\ \text{(empirical risk)}$$

10. The eight-stage end-to-end pipeline (Figure 5)

1. Problem formulation → 2. Data → 3. Representation → 4. Model → 5. Evaluation → 6. Decision → 7. Deployment → 8. Monitoring, with continuous feedback from monitoring back to every earlier stage. It is "not a rigid, one-directional waterfall"; it is iterative and cyclical.

11. Problem formulation essentials

A robust problem statement identifies: the target population and the decision-maker; the specific decision and the exact time it occurs; the desired outcome and the alternatives (including the baseline or status quo); the consequences of errors (FP vs FN); and the operational and ethical constraints.

> **From the notes:** Poor framing: "Build an AI model to predict hospital readmission." Better framing: "At the time of discharge, estimate the probability of an unplanned 30-day readmission among adult medical

> patients so that the discharge team can allocate a limited number of follow-up appointments, while measuring calibration, subgroup performance, preventability, and the consequences of false alerts."

Problem-framing worksheet (Figure 6) — memorize this sentence:

> **From the notes:** At [decision time], for [target population or unit], using only [available information], produce [required output] concerning [prediction horizon or outcome period], so that [decision-maker] can [action]. The system will be compared with [baseline] and evaluated using [metrics], while accounting for [error costs, subgroup risks, operational constraints, and governance requirements].

Problem-framing canvas (1.7.2): The Decision; The Actor; The Actions; The Necessity of AI; The Simpler Alternatives.

Begin with the decision, not the dataset (1.7.1): Identify the Decision → Determine Information Needs → Select the Analytical Paradigm (prediction, causal inference, optimization, automation) → Define the Operating Context → Source Purpose-Built Data → Evaluate Against the Decision.

12. Data questions and artifacts (1.5.2)

Critical questions: **Provenance and purpose; Definitions and proxies; Completeness and consistency; Temporal and relational integrity; Governance.**

Typical data artifacts: raw data inventories and source-system documentation; data dictionaries and label specifications; provenance records and quality reports; consent, licensing and access-control documentation; defined train-validation-test partitions; dataset datasheets.

Datasheet for Datasets documents: motivation, composition, collection methods, preprocessing steps, recommended uses, and limitations (skin-lesion example).

13. Six representation risks (1.5.3)

Risk	Definition and sepsis example
Loss of context	Reducing a dynamic entity to a static vector strips temporal progression. A 24-hour average heart rate hides a spike one hour ago.
Proxy encoding	Neutral-looking features (postal code, IP address) encode protected attributes. Zip code acts as a proxy for socioeconomic status and race.
Aggregation bias	Averages hide subgroup disparities. 92% hospital-wide accuracy hides failure for children and autoimmune patients.
Temporal leakage	A feature recorded after the prediction moment. Antibiotic dose counts updated after diagnosis give 99% test accuracy but fail live.
Identity errors	Records wrongly merged or one person split into several IDs. A transferred patient gets a temporary ID.
Train-serving mismatch	Different preprocessing in training vs live inference. Missing BP imputed with ward median in training but 0 in production, causing floods of false alarms.

14. Model selection criteria (1.5.4)

$$f_\theta : X \to Y$$

Criteria: **data characteristics** (size, dimensionality, nonlinearity); **task requirements** (classification, regression, ranking, need for calibration); **operational constraints** (latency, memory, compute); **governance and trust** (interpretability, explainability, robustness); **lifecycle costs** (maintenance burden, available expertise). "The most complex model is not automatically the best model." Occam's principle (parsimony) is a heuristic, not a law (TB plus rheumatoid arthritis example).

15. Evaluation (1.5.5)

Four dimensions: **Predictive performance** (discrimination and calibration); **Fairness and robustness** (subgroup performance, robustness, interpretability); **Operational viability** (cost, latency, stability); **Real-world impact** (decision utility, comparison with meaningful baselines, human-system performance).

Claim	Required validation
Temporal	Temporal validation: train on past, test on strictly future data (not random shuffled CV).
Geographic	External validation on data from entirely different sites.
Equity	Explicit subgroup performance evaluation.
Probabilistic	Calibration assessment (calibration curve), not just ranking.
Decision	Assessment within the actual workflow (simulation or prospective trial).

Model Card records: intended use, evaluation procedures, conditions under which it performs well, detailed subgroup results, and known limitations.

16. Decision stage (1.5.6)

$$\hat{p} = \hat{P}(Y = 1 \mid X = x)\ a(x) = \text{intervene if } \hat{p} \geq t\text{; do not intervene if } \hat{p} < t$$

The threshold t is never a default 0.5. It reflects: **asymmetric error costs; intervention capacity; uncertainty and expected benefit; fairness requirements; human-in-the-loop.**

$$C(t) = c_FP \cdot P(FP|t) + c_FN \cdot P(FN|t) + c_delay \cdot E[D|t] + c_workflow(t)$$

"The optimal decision threshold is rarely the one that simply maximizes overall accuracy." (Baggage screening: accuracy-maximizing t = 0.4 breaks the checkpoint; cost-minimizing t = 0.7.)

17. Deployment questions (1.5.7)

- **Infrastructure and performance:** where inference occurs (cloud, edge, on-premise); maximum latency; which features are available in real time.
- **Resilience and failure handling:** protocol for missing, delayed or corrupted inputs; abstention; fallback procedure; instant rollback.
- **Human-system interaction:** who receives the output; how reasoning and uncertainty are explained.
- **Governance and MLOps:** version control of models, code and pipelines; who holds accountability for monitoring and retraining.

"A highly accurate model sitting in a Jupyter notebook is not a deployed AI system."

18. Monitoring dimensions (1.5.8)

Input quality; input distribution (data drift); concept and label drift; feature pipelines; model outputs; predictive performance; ground-truth outcomes; operational health; human-system interaction (override rates, automation bias); harm, fairness and safety; feedback loops.

Hidden technical debt (Sculley): undeclared consumers and data dependencies; self-reinforcing feedback loops; system entanglement and configuration complexity; glue code; plus stale features, rigid UI categories and silent failures. Remedy: multi-dimensional monitoring, schema validation, counterfactual logging, a centralized version-controlled feature store.

19. Types of AI problems (Table 2)

Type	Output	Typical evaluation	Common framing error
Classification	Discrete class	Accuracy, sensitivity, specificity, F1, AUROC, AUPR	Treating all errors as equally costly
Regression	Continuous value	MAE, RMSE, R^2, interval coverage	Ignoring skew, heteroscedasticity or asymmetric cost
Ranking	Ordered items	NDCG, MAP, precision@k	Evaluating items independently of list position
Recommendation	Personalized items	Recall@k, NDCG, utility, diversity	Optimizing clicks rather than long-term benefit
Clustering	Unlabeled groups	Silhouette, stability, domain validity	Treating clusters as objectively real categories
Search	State, path or document	Relevance, path cost, success rate, latency	Defining an incomplete search space
Planning	Sequence of actions	Goal success, cumulative cost, constraint violations	Assuming a static or fully known environment
Control	Sequential feedback action	Stability, tracking error, safety, energy	Ignoring delayed or noisy feedback
Anomaly detection	Anomaly score	Precision-recall, detection delay, false-alarm rate	Assuming all rare cases are harmful
Generation	New content	Human evaluation, factuality, task success, safety	Equating fluency with truth
Forecasting	Time-indexed future values	MAE, RMSE, MASE, interval coverage	Randomly splitting temporally ordered data
Causal estimation	Effect of an intervention	Bias, interval coverage, sensitivity analysis	Interpreting correlation as intervention effect
Decision optimization	Feasible action	Expected utility, regret, constraint satisfaction	Optimizing an inaccurate or incomplete objective

Formulas: binary $Y \in \{0,1\}$; multiclass $Y \in \{1,\dots,K\}$; multilabel $Y \subseteq \{1,\dots,K\}$; regression $Y \in \mathbb{R}$; control $x_{t+1} = F(x_t, u_t, w_t)$; forecasting $\hat{y}_{t+h|t} = E[Y_{t+h} \mid I_t]$; causal effect $\tau = E[Y(1) - Y(0)]$; decision $a^* = \operatorname{argmax}_{a \in A} E[U(a,Y) \mid X = x]$.

20. The framing elements (Section 1.7)

Element	What it means
Unit of analysis	The independent entity being predicted for: an individual, an event, an object, or a spatial entity. Mixing the same unit across train and test leaks (chest X-ray example).
Index time	The exact moment the prediction is made. Every feature must be available at or before it.
Prediction horizon	How far ahead the system looks. Must match the operational window to act (wind turbine: 24 h useless, 14 to 30 days useful).
Inputs	Specify definition and source; unit and type; update frequency and missingness; timestamp semantics; availability at inference; privacy classification; transformation; known biases.
Output	Class, value, probability, interval, ranking, anomaly score, risk tier, recommendation, plan, action, generated artifact, or abstention. Probabilities beat hard labels.
Labels	Sources: objective measurement; human evaluation; system and administrative records; derived proxies. Interrogate: construct validity; access bias; measurement disparities; meaning of "negative"; status quo bias.
Constraints	Technical and infrastructure; resource and operational; trust and usability; governance, ethics and compliance.

Element	What it means
Stakeholders	Subjects; targets; operators and experts; broader ecosystem. Objectives often conflict.
Cost of error	TP (hit), FN (miss), FP (false alarm), TN (correct rejection); costs are asymmetric and include non-financial harms.
Baseline	Current professional judgment; a simple heuristic; the status quo; a legacy system; alternative policies.
Evaluation plan	Metrics (primary and guardrail); validation strategy; reliability and fairness; operationalization and governance. Locked before training to prevent metric shopping and data dredging.

21. Eight failure modes (Section 1.9)

1) Solving the wrong problem; 2) Poor data quality; 3) Data leakage (target, temporal, group, preprocessing, label, benchmark contamination); 4) Weak evaluation; 5) Underspecification; 6) Distribution shift; 7) Ignoring workflow and human factors; 8) No deployment or monitoring plan.

22. Data engineering (Section 1.10)

Data lifecycle (Figure 8): **generated → captured → validated → stored → transformed → served → observed → governed.** Data engineering must engineer semantics, temporal correctness (t_feature availability ≤ t), training-serving consistency, lineage, scale and timeliness, governance in infrastructure, and monitoring data. "Reliable model ⇏ Reliable AI system." Data engineering is "the connective tissue of the complete AI lifecycle."

23. Ten principles of sound AI problem formulation (Figure 9)

1) Begin with a decision or capability, not an algorithm. 2) Define the unit of analysis, population, index time and horizon explicitly. 3) Use only information available when the decision must be made. 4) Treat labels as measurements with their own error-generating processes. 5) Match evaluation to the claimed deployment environment. 6) Evaluate probabilities, decisions, subgroups and operations, not only average prediction. 7) Compare against a meaningful baseline. 8) Design deployment and monitoring before declaring model success. 9) Document data, model and governance assumptions. 10) Treat AI as a socio-technical system whose effects depend on people and institutions.

24. Useful metric formulas

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad \text{Recall (sensitivity)} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1} = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}) \quad \text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$$

$$\text{Alert volume} = \text{Number of eligible encounters} \times \text{P}(\hat{p} \geq t)$$

- **Calibration:** predicted probabilities match observed frequencies (a 30% forecast means rain on 3 of 10 such days).
- **Discrimination:** the ability to rank or separate positives from negatives (AUROC).
- **Precision@k / Recall@k:** used when capacity is fixed (for example 100 appointment slots or 300 bootcamp seats).

Part 3: Your Assignment (Report and Notebook) Recap

Sir said one quiz question will come from the report or the notebook. This part summarizes what you submitted and gives ready answers to the questions most likely to be asked. Learn the numbers and the reasoning, not just the conclusions.

3.1 Report overview

Problem	What you produced	Core concepts tested
1. RYDA problem framing	Completed the Figure 6 worksheet for allocating 300 bootcamp seats; justified index time, leakage, outcome definition, capacity rule, error costs and subgroups.	Index time, horizon, temporal leakage, label definition, ranking under capacity (precision@k), C(t), subgroup analysis
2. WePrep Model Card	Model card for a multimodal mock-interview emotion and "Interview Readiness" system; recommended Conditionally Approve (self-practice only).	Model card sections, subgroup performance, calibration, construct validity, appropriate vs prohibited uses, monitoring triggers
3. Crop disease Datasheet	Datasheet for an 80,000-image leaf dataset; concluded it is not suitable on its own for smallholders on low-cost phones.	Datasheet sections, selection bias, proxies and shortcuts, device shift, label verification, versioning
4. Résumé screening failure (2014 to 2015)	End-to-end failure analysis of the automated hiring tool; listed mandatory safeguards for a controlled pilot.	Status quo bias, proxy encoding, feedback loops, technical debt, automation bias, procedural fairness, NIST functions

3.2 Likely questions from Problem 1 (RYDA worksheet)

> **Why did you choose "end of orientation" as the index time rather than registration or the three-month mark?**

At registration the record is thin (no aptitude score) and the population is unstable, and early registration depends on internet access (the source of the first-come bias). At three months the model would be more accurate, but half of the 180-day window is gone; as the notes say about sepsis, a prediction made after the problem is visible "may be accurate but operationally unhelpful". End of orientation is the only point where all pre-intervention features exist and almost the whole horizon remains, so the horizon matches the window to act.

> **Which variables would cause temporal leakage in RYDA?**

Interview callbacks during the job search (future information and almost a direct measure of the outcome); verified employment at six months (the label itself, target leakage); anything created after orientation (support tier received, module completion, workshop attendance, three-month status); and fields edited after the fact (updated education or skills, aptitude score recorded late). The rule: every input must be known at or before the index time, and timestamp semantics (event time vs recording time) must be checked.

> **How did you make "high risk of remaining unemployed" measurable?**

Event: Y = 1 if the candidate is not in verified employment at day 180 (paid work of at least 20 hours a week, lasting at least four consecutive weeks, confirmed by a document). Formal, temporary, apprenticeship and freelance work count; one-off gigs and unpaid internships do not; full-time education is a separate outcome; unreachable candidates are recorded as missing, not coded 0 (the meaning of "negative"). High risk: inside the

top 300 by $\hat{p}$ and above a minimum floor t. Label checks: construct validity, measurement disparity (informal rural work is harder to verify), status quo bias, and the fact that high risk does not prove the bootcamp will help (a causal question).

Why is the output a ranking with k = 300 rather than a binary classifier?

With a fixed 0.5 threshold the number of positives changes every cycle (it might flag 700 people for 300 seats, or fewer than 300 and leave seats empty). The capacity constraint turns the problem into ranking under a fixed budget, evaluated with Precision@300 and Recall@300; the effective threshold is the $\hat{p}$ of the 300th candidate. Calibration still matters for assigning lower tiers, showing unmet need, and judging whether the 300th and 320th candidates are really different. Borderline and incomplete records go to human review (abstention).

What are the consequences of false positives and false negatives in RYDA?

A false positive spends an expensive seat on someone who would have found work anyway, and because seats are fixed it displaces someone who needed it. A false negative leaves a vulnerable candidate with only job alerts: six more months unsupported, lost income, and a risk of long-term unemployment, which is harder to reverse. Therefore $c_{FN} > c_{FP}$, Recall@300 is the primary metric, and subgroups (rural vs urban, income, gender, education, intersections such as rural × female) are reported separately with sample sizes and confidence intervals.

3.3 Likely questions from Problem 2 (WePrep Model Card)

What did the WePrep subgroup results show, and why does 84% overall accuracy mislead?

Native English speakers 89% vs non-native 62% (27-point gap); lighter skin 87% vs darker 68% (19 points); glasses or low light 55%. The aggregate hides these gaps (aggregation bias). Errors from the speech-to-text branch and the facial branch compound for users who belong to several groups, and intersectional results were not reported.

Why is calibration a concern for WePrep even though no calibration was reported?

A CNN trained mostly on well-lit, lighter-skinned faces still produces confident outputs on dark, poorly lit or partly covered faces, because nothing tells it the input is out of distribution. Confidence stays high while accuracy falls to 55 to 68%, so the gap between confidence and accuracy is widest for the worst-served groups. The 0 to 100 Readiness score makes it worse by compressing two uncertain outputs into one precise-looking number. Remedy: reliability diagrams per subgroup and abstention ("conditions insufficient for feedback").

What was your deployment recommendation and why?

Conditionally approve, for low-stakes voluntary self-practice only: hide the numeric score until the worst subgroups reach an accuracy floor, disclose accuracy gaps before recording, make scores impossible to share with employers, abstain on low-quality input, provide a correction channel, and monitor per subgroup. Any hiring or assessment use is not approved. This mirrors the notes: "information is not automatically meaning" (a smile can signal politeness or nervousness, the facial-emotion example).

3.4 Likely questions from Problem 3 (Crop disease Datasheet)

Why is the dataset not suitable for smallholder farmers using low-cost phones?

Every factor deployment depends on is missing: no smallholder farms; no drought images (drought stress can look like disease); 95% high-resolution camera images vs a phone deployment (train-serving mismatch and device shift); 70% of images from two regions; 90% of labels unverified by pathologists; healthy plants under-sampled (pushes the model toward over-diagnosis); and farm backgrounds available as shortcuts. The model would be "confidently wrong rather than obviously broken", and the cost falls on the farmers least able to absorb it. It is still useful for pre-training, benchmarking and research on shift and label noise.

What shortcuts or proxies could a model learn from this dataset?

Background (soil, mulch, staking, irrigation equipment differ by farm and farms differ in disease); device characteristics; photographer framing (lesions framed closely and centrally); lighting and time of day; and resizing artifacts. Splits should be made by farm code (or plot) to avoid group leakage.

3.5 Likely questions from Problem 4 (Résumé screening failure)

How did the model learn to prefer men without a gender field?

The label was ten years of past hiring decisions, which records what past recruiters did (status quo bias), not candidate quality. In a male-dominated corpus the model learned what a historically successful résumé looks like, and many features correlate with gender (women's clubs, women's colleges, verbs, career gaps). A high-capacity model combines these weak signals and effectively reconstructs gender (proxy encoding).

Why was "neutralizing" specific words flawed, and what technical debt did it create?

It treats the symptom, not the cause (the label); it can never be shown to be complete; it hides the evidence, turning a visible failure into a silent one; and the list grows with every retraining. It creates hidden post-processing and configuration debt (an unversioned, ownerless list outside the model), entanglement (changing one term shifts the whole decision surface), and a monitoring gap.

Describe the feedback loop in the hiring tool.

Biased labels in → the ranking decides who is interviewed (biased exposure) → low-ranked women are never observed (selective labels) → new hires chosen under the model's influence become fresh training data → the model appears more accurate because it is evaluated on people it selected (self-confirmation).

What safeguards are mandatory for a controlled pilot?

Fix the label first (a validated, job-relevant target) and pre-register the evaluation with a stop rule; staged deployment (silent/shadow mode, then advisory with a comparison group, with rollback ready); interface controls (show the full pool with confidence and explanations, never auto-reject, log overrides with reasons); continuous monitoring and release gates (subgroup selection rates, counterfactual logging, automated model validation); procedural safeguards (notice, appeal, audit, a model card, a named accountable owner, NIST Govern-Map-Measure-Manage, and a sunset condition).

3.6 Notebook (PA1) — worked answers

Problem 1: Forward pass of an MLP (one hidden layer)

Layer rule: z = W·x + b. Hidden activation: ReLU, f(x) = max(0, x). Output activation: sigmoid, σ(x) = 1 / (1 + e^(−x)).

Input x = [0.5, −1.2, 3.0]. Hidden layer (4 neurons), b1 = [0.1, −0.1, 0.2, 0.0]:

Neuron	W1 row · x + b1	z1	a1 = ReLU(z1)
1	0.2(0.5) + (−0.5)(−1.2) + 0.1(3.0) + 0.1 = 0.1 + 0.6 + 0.3 + 0.1	1.10	1.10
2	(−0.4)(0.5) + 0.8(−1.2) + (−0.2)(3.0) − 0.1 = −0.2 − 0.96 − 0.6 − 0.1	−1.86	0
3	0.6(0.5) + 0.1(−1.2) + (−0.3)(3.0) + 0.2 = 0.3 − 0.12 − 0.9 + 0.2	−0.52	0
4	(−0.1)(0.5) + (−0.2)(−1.2) + 0.9(3.0) + 0.0 = −0.05 + 0.24 + 2.7	2.89	2.89

Output layer: W2 = [0.5, −0.6, 0.3, −0.1], b2 = −0.2.

$$z2 = 0.5(1.1) - 0.6(0) + 0.3(0) - 0.1(2.89) - 0.2 = 0.55 - 0.289 - 0.2 = 0.061$$

$$\text{final output} = \sigma(0.061) = 1 \,/\, (1 + e\text{^}(-0.061)) \approx 0.5152$$

Matches the notebook output: a1 = [1.1, 0, 0, 2.89], final output = 0.51524527.

> **Why is ReLU used in the hidden layer and sigmoid at the output?**

ReLU adds non-linearity cheaply (without it, stacked linear layers collapse into one linear map) and zeroes negative pre-activations (neurons 2 and 3 above are "switched off"). Sigmoid squashes the final value into (0, 1) so it can be read as a probability p̂. Link to the notes: an MLP is a deep learning model $f_\theta: X \to Y$ whose parameters θ are the weights and biases; a probability output is preferable to a hard label because it keeps uncertainty and lets the threshold t be set by error costs and capacity. Note that sigmoid output is not automatically calibrated.

Problem 2: Bayes' theorem (spam given "urgent")

$$P(\text{Spam}) = 2000/10000 = 0.2 \; P(\text{Not Spam}) = 8000/10000 = 0.8$$

$$P(\text{"urgent"}) = 0.80 \times 0.2 + 0.05 \times 0.8 = 0.16 + 0.04 = 0.20 \text{ (law of total probability)}$$

$$P(\text{Spam} \mid \text{"urgent"}) = (0.80 \times 0.2) \,/\, 0.20 = 0.16 \,/\, 0.20 = 0.80$$

> **Interpret the Bayes result. Why is the posterior (0.8) much higher than the prior (0.2)?**

The word "urgent" is 16 times more likely in spam (0.80) than in non-spam (0.05), so observing it strongly updates the belief toward spam. This is the notes' view of probabilistic intelligence: "learning uncertain relationships from evidence; updating beliefs when new information becomes available; quantifying confidence". Note the base rate still matters: even with a strong signal, 20% of emails containing "urgent" are legitimate, so auto-deleting them would cause false positives (quarantining a critical client email is costly, per the notes' asymmetric-cost example). Naive Bayes extends this by multiplying likelihoods of many words under a conditional-independence assumption.

Problem 3: List vs set lookup (algorithmic complexity)

Vocabulary size N	List time (s)	Set time (s)	List/Set ratio
10,000	0.000163	0.0000027	~60×
50,000	0.000967	0.0000032	~300×
100,000	0.002044	0.0000034	~600×

Vocabulary size N	List time (s)	Set time (s)	List/Set ratio
250,000	0.005821	0.0000041	~1,400×
500,000	0.009852	0.0000031	~3,200×

How do the curves confirm O(N) for a list and O(1) for a set?

The target is the last token, so the list must compare against every element: time grows roughly in proportion to N (500,000 costs about 60 times 10,000; a straight line through the origin). The set computes one hash and probes one bucket, so its time stays flat near the timer's resolution regardless of N.

What is the trade-off of a hash set in a memory-constrained production environment?

Memory and construction cost. The hash table is kept deliberately sparse to limit collisions, so it takes several times the list's memory, and building it requires hashing every token once. On a memory-limited edge device, a sorted array with binary search (O(log N)) may be preferable. Link to the notes: this is exactly the **operational constraints** criterion of model and system selection (latency, memory footprint, computational limits) and the **technical and infrastructure constraints** of Section 1.7.9; lower latency is not free.

Part 4: Practice Questions 1 to 50

Each answer uses the notes' definitions and examples. In the quiz, a strong short answer = definition + reason + example from the notes.

Historical and Philosophical Roots of AI

Q1. Why is the history of AI better understood as a history of attempts to formalize intelligence rather than only as a history of faster computers?

Long before neural networks and GPUs, mathematicians, philosophers, engineers, psychologists, linguists and economists asked whether intelligence could be understood in formal terms, as rules, symbols, probabilities, algorithms or mathematical models. Each attempt to formalize intelligence (logic, probability, computation, information theory, feedback, cognitive science, optimization) contributed concepts that still shape AI even though the hardware changed. Faster computers only made it possible to run these formalizations; the core question was always: "Can perception, reasoning, learning, decision-making, and intelligent action be represented sufficiently precisely to be carried out by a machine?" So AI is "not a single invention" but a convergence of traditions.

Q2. Identify the major intellectual traditions that contributed to the development of modern AI.

- formal logic and the mechanization of reasoning;
- probability and statistical inference;
- computation and algorithms;
- information theory (Shannon);
- control and feedback (Wiener's cybernetics);
- cognitive science and theories of intelligence;
- operations research and optimization;
- philosophy of mind, language, knowledge and agency.

Q3. Explain how formal logic made the idea of mechanized reasoning possible.

Formal logic represents valid reasoning through formal symbols and rules. Once an argument is formalized, each step follows from explicit rules, so one can ask whether the steps can be executed mechanically without understanding. This idea led to symbolic AI, theorem proving, planning, expert systems, knowledge representation and logic programming. Its underlying position is that some forms of intelligence are "operations over structured representations", visible today in search over state spaces, planners applying operators, knowledge graphs and rule engines. Prolog is the classic example: it derives grandparent(tom, ann) from facts and a rule by mechanical proof.

Q4. What does it mean to say that intelligence requires representation?

A machine "cannot reason directly about an unrepresented world". It can only reason over constructed representations: variables, symbols, vectors, tokens, states, graphs or probabilities. Whatever is not represented is invisible to it. Representation therefore decides which patterns and distinctions a system can learn or reason about (the same idea returns in Section 1.5.3, where representation is "the bridge between the physical world and computational models"). Related idea: a world model is an internal representation of how an environment works that lets an agent predict and plan.

Q5. How does a symbolic program differ from a numerical program?

A numerical program (a calculator or a neural network) mainly manipulates continuous numbers and statistical weights. A symbolic program manipulates discrete symbols, logical structures and explicit rules to perform reasoning. It "does not understand the physical world; it only understands the representations it has been given." Example: a normal math library computes 2.5 × 2.5 = 6.25, while SymPy manipulates the symbols x and y themselves to prove that $(x + y)^2$ equals $x^2 + 2xy + y^2$.

Q6. Explain how facts, rules, and queries interact in a Prolog program.

- **Facts (knowledge representation):** parent(tom, bob). parent(tom, liz). parent(bob, ann).
- **Rules (formal logic):** grandparent(X, Y) :- parent(X, Z), parent(Z, Y). The symbol :- means "is true if".
- **Query (mechanized reasoning):** ?- grandparent(tom, ann). asks "can you prove this?"

Prolog has no stored grandparent(tom, ann) fact. It searches its formal state space, matches the query against the rule, finds Z = bob that satisfies both parent facts, and mechanically deduces True. It does not know who Tom or Ann are.

Q7. How does an expert system separate knowledge representation from the inference engine?

The knowledge (the rules) is stored separately from the reasoning engine (the code that applies them). In the notes' Python example, patient symptoms are stored as a set {"fever", "cough", "fatigue"}, and the medical rules are a list of if-then entries written by humans (for example {fever, cough, fatigue} → Flu). The inference engine, diagnose(), loops over the rules and checks whether each rule's conditions are a subset of the symptoms, returning the first match (output: Flu). Because they are separate, experts can change rules without changing the engine. The computer "does not discover these relationships. It simply follows them."

Q8. How does symbolic theorem proving establish that two algebraic expressions are equivalent?

SymPy defines abstract symbols (x, y), builds symbolic expressions expr1 = $(x + y)^2$ and expr2 = $x^2 + 2xy + y^2$, and then mechanically simplifies their difference using formal algebraic rules (such as the distributive property). Because simplify(expr1 − expr2) reduces to the canonical form 0, the system has mechanically proven the expressions are equivalent. It does not understand the theorem like a mathematician; it has rules for manipulating expressions and checks whether both sides reduce to the same canonical form.

Q9. Why is symbolic reasoning bounded by the symbols, facts, and rules supplied to the system?

The symbolic machine does not learn patterns from data; it only applies the rules it was given to the representations it was given. "If you do not provide the symbol, the fact, or the rule, the machine cannot reason about it." For example, the expert system returns "Unknown condition" for any symptom combination not in its rule list, and Prolog cannot prove a relationship it has no rule for. This is the lesson that "reasoning is just computation over structured representations", and it explains brittleness and the difficulty of acquiring and maintaining knowledge.

Q10. Compare the roles of probability theory and statistical inference in reasoning under uncertainty.

- **Probability theory** provides the language for uncertainty. It starts from a known model or assumptions and works forward to the likelihood of outcomes (for example, "this patient has an 80% chance of the condition given the evidence").
- **Statistical inference** works in the opposite direction: from observed sample data to conclusions about an unknown population or parameters (for example, does this test reliably predict disease across thousands of patients?). It also separates observed patterns in a sample from population-level claims.

Together they changed the idea of intelligence from applying known rules to learning uncertain relationships from evidence, updating beliefs, quantifying confidence, and selecting actions under risk. Modern ML inherits this: parameters are estimated from data and predictions are "uncertain statements rather than infallible facts".

Feedback, Information, and the Operational View of Intelligence

Q11. How did cybernetics connect communication, feedback, and control in animals and machines?

Norbert Wiener's Cybernetics (1948) studied control and communication "in the animal and the machine". In a feedback system, the consequences of previous actions are observed (communication of information) and used to modify future behaviour (control). This treated intelligent systems as dynamic, adaptive and embedded in environments rather than isolated calculators. Examples: a thermostat correcting temperature, a robot adjusting its movement, a recommender learning from interactions, a fraud system changing thresholds, a reinforcement-learning agent adapting its policy through rewards.

Q12. Explain how a feedback loop can make an AI system an active participant in the environment that it models.

Once predictions influence decisions, those decisions change the world, and the changed world produces the next training data. The data then reflect not just reality but also the system's earlier choices. In the predictive policing example, the model predicts high crime in an area, more police are sent, more minor incidents are recorded, and the new data "confirm" the prediction. "The model has changed the environment it is trying to predict." So AI systems are not passive observers; good design must consider how predictions reshape future data and outcomes.

Q13. How can predictive policing produce self-reinforcing bias and selective labels?

- Historical data suggest a neighbourhood is high risk.
- The system sends more patrols there.
- More observation produces more recorded incidents.
- The new data reinforce the original prediction, and the model becomes more confident.

Self-reinforcing bias: the historical pattern is amplified each cycle. **Selective labels:** crime is only recorded where police were sent to look, so the data do not show crime in unpatrolled areas; the labels reflect where attention went, not the true underlying crime rate.

Q14. Distinguish among self-reinforcing bias, selective labels, performative effects, and feedback loops.

Term	Meaning
Self-reinforcing bias	Existing patterns in historical data become amplified over time.
Selective labels	Outcomes are observed only where the system directed attention (or for people it selected), not for the full population.
Performative effects	The prediction itself changes behaviour and creates conditions that make it appear accurate.
Feedback loops	The output of one prediction becomes part of the input for future predictions, potentially locking the system into a harmful cycle.

The feedback loop is the mechanism; selective labels and performative effects are how it distorts the data; self-reinforcing bias is the result.

Q15. Why is information not automatically equivalent to meaning?

Shannon showed that information can be measured and processed mathematically, so images, speech and text can all be encoded as data. But "a dataset may contain billions of values while failing to represent the phenomenon of interest." Meaning depends on how variables were defined, measured, collected, labelled and interpreted. A pixel value does not inherently represent happiness. So a model can find precise statistical patterns in the information that are nonetheless misleading about the real world.

Q16. Use facial emotion recognition to explain how a dataset can contain abundant information but represent the target phenomenon poorly.

A face photo is millions of pixel values: plenty of information. But the meaning of the label depends on questions such as: How were emotions defined? Who labelled them? Were labels based on genuine feelings, facial expressions or human judgement? Did the data include diverse cultures, ages, lighting and contexts? If smiling faces are labelled "happy", the model learns smile = happiness, yet a smile can signal politeness, nervousness, discomfort or social expectation. The model can reach high accuracy on its dataset while misunderstanding human emotion. The three levels: information theory (pixels), machine learning (statistical relationships), meaning (what variables really represent).

Q17. How did Claude Shannon contribute to concepts now used throughout machine learning?

His 1948 paper, A Mathematical Theory of Communication, formalized information, uncertainty, coding and transmission. It supplied concepts used throughout ML: **entropy, cross-entropy (the standard classification loss), mutual information, compression, channel capacity and noisy communication.** Its biggest consequence was that information became measurable and manipulable, so any signal (images, speech, text, sensor and biological data) could be encoded and processed as data.

Q18. How did Alan Turing transform the question of machine thought into an operational evaluation problem?

In 1950 Turing replaced the vague question "Can machines think?" with an operational test based on observable interaction: can a machine's conversational performance become indistinguishable from a person's under specified conditions? This moved the focus from unverifiable metaphysical claims to measurable behaviour. AI evaluation is still operational: Can the system classify correctly? Solve a planning problem? Answer questions? Generate coherent text? Control a vehicle? Outperform a reference baseline?

Q19. Why does successful performance on a Turing-style test not settle whether a machine understands?

Behavioural success does not answer the deeper questions: Does the system understand? Is it merely reproducing statistical regularities (the "stochastic parrots" concern)? Does intelligence require consciousness? Can linguistic performance be separated from grounding in the world? Should human-like behaviour be the standard? If thinking is defined as conscious experience rather than information processing, no amount of performance proves thought (Searle's argument). The notes separate "engineering tests of capability" from "philosophical claims about mind and understanding".

Q20. What role did the Dartmouth proposal play in naming and consolidating the field of AI?

The 1955 proposal (McCarthy, Minsky, Rochester, Shannon) for the 1956 Dartmouth Summer Research Project used the term "artificial intelligence" and proposed studying whether learning and intelligence could be described precisely enough for machines to simulate them. Themes: language, abstraction, problem solving, self-improvement, neural mechanisms. It treated intelligence as decomposable into computational

components and helped consolidate, name and legitimize the research traditions: symbolic reasoning, search and planning, machine learning, NLP, computer vision, robotics, knowledge representation and automated theorem proving.

Symbolic AI, Statistical Learning, and Hybrid Systems

Q21. State the main strengths of symbolic AI.

Explicit representations; inspectable reasoning chains; incorporation of domain rules; compositional structures; strong performance when the state space and rules are well specified (theorem proving, game playing, planning, expert systems). Newell and Simon argued physical symbol systems are central to intelligent action.

Q22. State the main limitations of symbolic AI in complex and uncertain environments.

Difficulty acquiring and maintaining knowledge; brittleness under ambiguity; poor handling of noisy perception; combinatorial explosion; dependence on manually encoded representations.

Q23. Why would a purely symbolic self-driving system struggle with ambiguity, noisy perception, and combinatorial explosion?

Rules such as "IF light is green THEN proceed" seem sufficient, but:

- **Ambiguity:** a pedestrian on the curb looking at a phone may or may not cross; the system needs a crisp symbolic state ("crossing" / "not crossing") that reality does not provide.
- **Noisy perception:** rain, fog, glare or a partly hidden sign can make perception mislabel a stop sign as a speed-limit sign; reasoning then starts from wrong symbols and can produce dangerous decisions.
- **Combinatorial explosion:** combinations of road conditions, weather, traffic density, pedestrian behaviour and vehicle interactions grow exponentially; rules for every combination are infeasible.
- Also: thousands of rules are needed for construction zones and emergency vehicles, and concepts like "safe following distance" must be hand-defined.

Hence real vehicles are hybrid: deep learning for perception, rules for traffic law, safety and high-level planning.

Q24. State the main strengths of statistical machine learning.

Learning from high-dimensional observations; tolerance of noisy inputs; adaptation from examples; strong performance in perception, prediction and generation. The developer specifies a model class, objective, data representation and training procedure instead of programming every rule.

Q25. State the main limitations of statistical machine learning.

Dependence on data quality; limited causal interpretation; potential opacity; vulnerability to distribution shift; difficulty guaranteeing behaviour outside the observed data regime.

Q26. How can a pneumonia prediction model learn a hospital-specific shortcut rather than a medically meaningful pattern?

Severely ill patients are more likely to receive portable X-rays, so in the training data "portable X-ray" is strongly associated with pneumonia. The model can learn to predict from the image markers of the portable machine rather than from lung pathology. It predicts from correlation, not the medical cause (limited causal

interpretation). At 98% test accuracy it looks ready, but at another hospital with different equipment and practices the shortcut no longer holds.

Q27. What is distribution shift, and why can it reduce performance at a new hospital?

Distribution shift means the deployment data differ from the training data. A model trained on one hospital's X-ray machines, imaging protocols and patient demographics meets different machines, protocols and populations elsewhere; its learned relationships (including shortcuts) no longer match, so accuracy drops. Causes listed in 1.9.6: population changes, new devices, policy changes, seasonal effects, user adaptation, economic shocks, new clinical practice, adversarial behaviour, pipeline changes. Remedy: external validation and monitoring.

Q28. Explain why symbolic AI and statistical machine learning should not be treated as a permanent binary choice.

Each has complementary strengths and weaknesses: statistical learning handles noisy perception but lacks guarantees and causal understanding; symbolic methods give explicit, inspectable rules but are brittle. Modern systems combine them: learned perception with rule-based constraints, neural representations with search, language models with retrieval, prediction with optimization, learned policies with safety controllers, and ML recommendations with human review. "The practical question is which representations and computational mechanisms are appropriate for the task."

Q29. Describe how an autonomous warehouse robot can combine learned perception, search, optimization, safety rules, and human review.

- **Learned perception:** deep learning identifies shelves, packages, barcodes and people under varying light.
- **Rule-based constraints:** never exceed the speed limit, always stop for a person, never enter restricted zones.
- **Neural representations with search:** a network estimates traversable paths; a search algorithm finds the shortest safe route.
- **Language models with retrieval:** "where is item X?" is answered from the live inventory database.
- **Prediction with optimization:** forecast warehouse traffic and optimize routes to minimize delivery time and congestion.
- **Learned policy with safety controller:** an RL policy drives smoothly, but an independent controller overrides it when collision risk appears.
- **Human review:** for a damaged package the robot recommends discard or inspect; a supervisor decides.

A purely symbolic robot could not interpret noisy images; a purely statistical one might violate safety rules.

Q30. Under what conditions is a hybrid AI system preferable to a purely symbolic or purely statistical system?

When the task has both (a) noisy, high-dimensional perception or patterns that are hard to hand-code (use learning) and (b) hard constraints, safety rules, legal requirements, or known domain logic that must always hold (use rules/search/optimization), and (c) high-stakes or ambiguous cases where accountability requires human review. Also when data are limited or shift over time and rules provide guarantees outside the observed data regime. Examples: autonomous vehicles, warehouse robots, clinical decision support, legal drafting with retrieval.

Philosophical Questions, Agency, and Responsibility

Q31. Why is intelligence better viewed as a spectrum of capabilities than as a single monolithic property?

No single definition covers every legitimate AI system. Intelligence includes many functional capacities: reasoning correctly, learning from experience, solving unfamiliar problems, using language, forming plans, acting toward goals, adapting, understanding context, interacting socially and creating artifacts. A route planner, an anomaly detector, a surgical robot and a language model each show different forms of computational competence. So it is more practical to characterize AI by which capabilities a system demonstrates.

Q32. Compare prediction with causal reasoning.

- **Prediction** answers "What is likely to happen?" It exploits statistical associations, which may include shortcuts, proxies and dataset artifacts.
- **Causal reasoning** answers a counterfactual question: "What would happen if a specific intervention were applied?" It identifies cause-and-effect relationships and, following Pearl, requires explicit structural assumptions and mechanistic models, which cannot be derived from observational associations alone.

Accurate prediction identifies who is likely to succeed; causal reasoning determines which actions will actually improve success.

Q33. Why can accurate prediction fail to answer a question about the effect of an intervention?

A predictor learns correlations in historical data, and those correlations may be driven by confounders or proxies rather than by the variable we would change. Intervening on the variable breaks the historical relationship. For example, a model predicting disease risk from hospital administrative markers, or student success from socioeconomic proxies, says nothing about what happens if we change those factors. The causal structure "generally cannot be derived from observational associations alone without additional assumptions or interventions" (for example randomized experiments). Also, both potential outcomes Y(1) and Y(0) are never observed for the same unit.

Q34. Distinguish among counterfactual inquiry, structural assumptions, and mechanistic models.

Concept	Asks / specifies	Notes example
Counterfactual inquiry	"What if we changed something?"	Would the student have passed if they had attended all lectures? Would crop yields rise with drip irrigation?
Structural assumptions	Which variables directly influence others (the causal pathways).	Smoking increases lung-cancer risk; rainfall influences crop production; traffic volume affects travel time.
Mechanistic models	How the process actually works to produce outcomes.	Weather model; pharmacokinetic model; epidemic model; traffic-flow model.

Q35. Use student attendance and exam performance to explain the difference between correlation and intervention effect.

A model finds that high-attendance students are much more likely to pass, and predicts well. But attendance may be a proxy for motivation, prior knowledge, study habits or family support. If the university made every student attend every lecture (an intervention), grades would not necessarily rise by the amount the

correlation suggests, because disengaged students without prerequisites or study time would still struggle. Prediction: "Students with high attendance are likely to pass." Causal: "If we increase a student's attendance, how much will their probability of passing actually improve?"

Q36. What does it mean to describe a deployed AI system as a socio-technical composite?

A deployed AI system is not just an algorithm. Its outputs are interpreted by people, built into workflows, and bounded by laws, resources, norms and incentives. Figure 4 shows five layers: **Model, Data Infrastructure, Interface, Human Workflow, Institutional Rules.** The same model can produce very different outcomes depending on the other four layers.

Q37. Explain how data infrastructure, interface design, human workflow, and institutional rules can alter the impact of the same recruitment model.

- **Model:** learns patterns associated with successful applicants.
- **Data infrastructure:** training on past hiring decisions carries historical preferences for certain universities, locations or groups into the model.
- **Interface:** dashboards and ranked lists shape whether recruiters critically evaluate or simply accept the ranking.
- **Human workflow:** treating the score as final vs as a supporting recommendation produces different outcomes.
- **Institutional rules:** employment law, fairness policies and ethics standards decide how it is used and monitored.

"A technically accurate model may still create unfair hiring practices if the data, workflow, or governance mechanisms are poorly designed."

Q38. Why should moral agency not be attributed directly to an algorithm?

An AI model "does not possess intent, judgment, or moral agency". It does not choose its training data, define its objectives, curate its labels, or decide where it is deployed. Operational parameters such as thresholds, confidence scores, human-review triggers and escalation policies are configured by developers and stakeholders. Every output reflects "a chain of human and institutional decisions". Blaming the algorithm hides who is actually accountable, so responsibility must be assigned to people across the lifecycle.

Q39. How do upstream and downstream human decisions shape AI outcomes?

- **Upstream** (design, data collection, training): choosing training data, defining the objective (for example, maximize repayment), selecting features (location may act as a proxy), setting thresholds.
- **Downstream** (after deployment): how outputs are interpreted and acted on, whether loan officers treat recommendations as final, whether applicants can request review, how regulations monitor the system.

In the loan example the responsible questions are: Who selected the training data? Who defined success? Who approved deployment? Who monitored performance? What safeguards existed for affected people? So Responsible AI cannot be solved by "tweaking the code"; it needs governance across the whole lifecycle.

Q40. Explain the Govern, Map, Measure, and Manage functions of the NIST AI Risk Management Framework.

- **Govern:** establish the culture, accountability structures and policies to manage AI risk (the bank assigns responsibility and creates policies).

- **Map:** identify and contextualize the specific risks of the system in its environment (unfair outcomes, privacy, impact on applicants).
- **Measure:** assess, analyze and track the risks with quantitative and qualitative metrics (accuracy, bias across groups, security, explainability).
- **Manage:** prioritize and act on risks by mitigating, transferring or accepting them (improve data, adjust processes, add human review, discontinue unsafe uses).

Goal: trustworthy AI that is valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, and fair with harmful bias managed. Governance is cross-cutting, not a final checklist.

The AI and Data Science Ecosystem

Q41. Define artificial intelligence as both a scientific discipline and an engineering discipline.

Definition: "the scientific and engineering discipline concerned with constructing computational systems that perceive or receive information, form useful representations, reason or learn from evidence, and select actions or outputs in pursuit of specified objectives under uncertainty and constraints."

- **As a science**, AI studies: What is intelligence? How can knowledge be represented? How can a system infer conclusions from evidence? How can an agent learn? How should decisions be made under uncertainty? What resources does intelligent behaviour need?
- **As engineering**, it builds useful systems: recognizing objects, predicting risks, allocating resources, recommending content, detecting anomalies, generating content, planning routes, controlling machines, assisting decisions.

The distinction matters because a system can be useful without reproducing human cognition (a weather model need not think like a meteorologist).

Q42. Compare systems that think like humans, act like humans, think rationally, and act rationally.

- **Think like humans:** reproduce human cognitive processes (memory, reasoning, perception); judged by whether the mechanism resembles human cognition. Roots in cognitive psychology, neuroscience, linguistics, philosophy of mind.
- **Act like humans:** judged only by observable behaviour; the Turing Test.
- **Think rationally:** formalize correct reasoning through logic (theorem proving, expert systems, constraint reasoning); limited because real environments are rarely complete or deterministic, so probability and decision theory are needed.
- **Act rationally:** a rational agent chooses actions expected to advance its objectives given information and constraints; handles uncertainty, incomplete information, limited computation, competing objectives, delayed consequences and exploration vs exploitation. Most useful for modern AI.

Q43. What is a rational agent, and why does rational action not imply perfect action?

A rational agent "selects actions expected to advance its objectives, given its information and constraints." Rational means choosing the best action relative to what it knows and can compute, not the action that turns out best in hindsight. With uncertainty, incomplete information, limited computation, competing objectives and delayed consequences, even the best choice can have a bad outcome. "Rational action does not mean perfect action. It means selecting an appropriate action relative to the information, objectives, and resources available."

Q44. Distinguish among passive analytical systems, decision-support systems, semi-autonomous systems, and autonomous systems.

- **Passive analytical:** turns data into insights or probabilities but cannot act (sales-forecast dashboard, disease-probability model, equipment-failure predictor).
- **Decision-support:** proposes specific actions; final authority and execution stay with humans (specialist-referral tool, ranking loans for underwriter review, maintenance priorities).
- **Semi-autonomous:** executes automatically in routine low-risk cases; escalates ambiguous or high-stakes cases (fraud blocking with borderline cases to investigators; screening that clears low-risk patients).
- **Autonomous:** perceives, decides and acts with little or no human intervention (manufacturing robots, warehouse sorting, high-frequency trading, closed-loop process control).

Q45. Why should autonomy not be confused with intelligence?

Autonomy is about the authority and mechanism to act without human involvement; intelligence is about the capability to perceive, reason and learn. A thermostat acts autonomously in a narrow loop but has no general intelligence. A powerful language model may show strong linguistic reasoning but has no agency: it cannot act on the physical or institutional world without human mediation. So high autonomy does not imply intelligence, and high intelligence does not require autonomy. The level of autonomy should be set by risk, not by how clever the system is.

Q46. Differentiate artificial intelligence, machine learning, data science, statistics, analytics, optimization, automation, generative AI, and data engineering.

Use Table 1 (Part 2, section 9). In one line each: **AI** is the broad umbrella for machines performing perception, reasoning, learning, planning, language, decision-making or adaptive action. **ML** is the major approach within AI that learns patterns or parameters from data. **Data science** is the broader lifecycle of acquiring, preparing, analyzing, modeling and communicating data, which may or may not use AI. **Statistics** is the inferential foundation: what can be inferred about a population from uncertain observations. **Analytics** is decision-oriented use of data (what happened, why, what may happen, what to do). **Optimization** finds the feasible choice that minimizes or maximizes an objective (used in training, planning, allocation, control). **Automation** executes a predefined process with limited human intervention and may contain no intelligence. **Generative AI** produces new content and is one family of AI capabilities, not all of AI. **Data engineering** builds the pipelines, schemas, stores, lineage and contracts on which data-driven AI depends.

Q47. Why is machine learning not synonymous with artificial intelligence?

ML is only one subset of AI. Many robust, high-stakes AI systems operate without learning from data, using: search algorithms (state spaces, game trees); formal logic and automated reasoning; constraint satisfaction and mathematical optimization; automated planning and scheduling; knowledge bases and expert systems; manually engineered control rules. "Machine learning is merely one powerful methodology for achieving intelligent behavior, not the exclusive definition."

Q48. Why is deep learning not synonymous with machine learning?

Deep learning is a specific class of ML defined by multi-layer artificial neural networks. Much of ML does not use deep networks: linear regression and kernel machines, decision trees and random forests, nearest-neighbour methods, Bayesian probabilistic models. "All deep learning is machine learning, [but] the vast majority of machine learning algorithms are not deep learning." (Your notebook's MLP is deep learning; your Bayes spam calculation is not.)

Q49. Why is data science broader than model training?

Data science covers analytical, investigative and strategic activities. Its outputs include causal estimates and experimental designs (A/B tests, RCTs); exploratory analyses and data-quality investigations; statistical reports and visualizations; forecasting systems and operational ML models. "Model training is just one tool in the data scientist's toolkit"; the real scope is to extract, validate and communicate actionable knowledge from data.

Q50. Compare the traditional emphasis of statistics with the traditional emphasis of machine learning.

- **Statistics** is rooted in inference: understanding the data-generating process. It emphasizes sampling and population inference, quantifying uncertainty and checking assumptions, estimation, identifiability and hypothesis testing. Example: an air-filter programme is evaluated with treatment and control neighbourhoods, estimation of the reduction, a hypothesis test, and a 95% confidence interval (15%, CI 10% to 20%).
- **Machine learning** is rooted in prediction: accurately mapping inputs to outputs. It emphasizes predictive performance and generalization, high-dimensional inputs and computational scalability, learned representations and automated function approximation. Example: tumour detection in 3D MRI with millions of voxels across many GPUs.

Breiman called these "the two cultures"; a mature practitioner combines both.

Part 5: Practice Questions 51 to 100

Q51. How do inferential thinking and predictive thinking complement each other in a mature AI project?

Predictive thinking (ML) delivers accurate mappings from inputs to outputs that generalize to unseen data. Inferential thinking (statistics) asks whether conclusions hold for the population, quantifies uncertainty, checks assumptions, and estimates effects of interventions. A mature project needs both: prediction to rank or flag cases, inference to check calibration and confidence intervals, test whether an intervention actually works (the air-filter study), and judge whether a sample represents the deployment population. Together they produce systems that are "not only highly accurate but also reliable, robust, and interpretable".

Q52. Why is mathematical optimization ethically blind and epistemically blind?

- **Ethically blind:** it has no moral agency; it cannot evaluate fairness, justice or harm (for example, minimizing hiring cost by discriminating, or maximizing engagement by amplifying misinformation).
- **Epistemically blind:** it cannot judge whether its data and assumptions are valid: whether y is the correct target, whether the sample represents the deployment population, whether the loss reflects real harm, or whether the model should be deployed.

It just minimizes the empirical risk $\hat{R}(\theta) = (1/n)\ \Sigma\ L(f_\theta(x_i), y_i)$ it is given.

Q53. Explain why optimization can solve a specified objective without validating whether the objective is appropriate.

The optimizer only sees the loss function and data; it does not know what the numbers mean. In the health-care example (Obermeyer et al.), the target was future health-care cost, assumed to be a proxy for health need. Because Black patients with the same illness historically generated lower costs (access barriers), the model learned they were lower risk and denied them care-management places. The optimization succeeded "brilliantly" at the specified objective while causing real harm. "Optimization solves the objective that was specified. It does not validate the objective." Validating the objective is the designers' responsibility.

Q54. Distinguish automation without intelligence from intelligence without full automation.

- **Automation without intelligence:** a deterministic rule executes a task mechanically without reasoning. Legacy fraud script: "if a transaction exceeds $10,000 and comes from a new IP address, freeze the account and send a rejection email". It cannot adapt to context, so a legitimate large purchase while travelling is blocked.
- **Intelligence without full automation:** an ML model analyzes hundreds of features (spending history, device fingerprint, geolocation, merchant risk) and outputs a nuanced assessment ("85% fraud, but matches this user's upcoming vacation"), then routes it to a human analyst who decides.

"Automation concerns how tasks are executed, whereas AI concerns how information is analyzed, patterns are inferred, and decisions are supported."

Q55. Compare generative AI with discriminative AI and identify the distinctive risks of generative systems.

- **Discriminative AI** analyzes existing data to classify, rank, score or predict (a model flags risky clauses in 50,000 contracts).
- **Generative AI** learns the underlying distribution of data and synthesizes new outputs: text, images, audio, code (an LLM drafts a new NDA).

NIST Generative AI Profile risks: **confabulation** (hallucinated citations to non-existent cases); **harmful content**; **information integrity** (deepfakes, or a professional-looking draft accepted without verification); **intellectual property and privacy** (reproducing confidential settlement terms from training data); **misuse** (jailbreaking to produce exploitative clauses). Law firms therefore use discriminative models for low-risk high-volume tasks and heavy governance with mandatory human review for generative drafting.

AI as an End-to-End Pipeline

Q56. List and explain the eight stages of the end-to-end AI pipeline presented in the module.

1. **Problem formulation:** define the real-world decision, population, decision-maker, timing, outcome, baseline, error costs and constraints.
2. **Data:** acquire and prepare data while interrogating provenance, definitions, completeness, temporal integrity and governance; document with datasheets.
3. **Representation:** translate real entities into feature vectors, tensors, tokens, graphs or sequences; avoid loss of context, proxies, aggregation bias, leakage, identity errors and train-serving mismatch.
4. **Model:** learn $f_\theta: X \rightarrow Y$, chosen by data characteristics, task requirements, operational constraints, governance and lifecycle cost.
5. **Evaluation:** assess predictive performance (discrimination and calibration), fairness and robustness, operational viability and real-world impact; match tests to deployment claims; document in a Model Card.
6. **Decision:** turn $\hat{p}$ into action with a threshold t set by error costs, capacity, uncertainty, fairness and human review; minimize $C(t)$.
7. **Deployment:** embed into live infrastructure with latency limits, fallbacks, abstention, rollback, explanations and MLOps governance.
8. **Monitoring:** watch inputs, drift, pipelines, outputs, performance, operations, human interaction, fairness and feedback loops.

Q57. Why should the AI pipeline be viewed as iterative and cyclical rather than as a one-directional waterfall?

Insights from later stages, especially monitoring, often reveal flawed assumptions, data drift or performance degradation. These trigger feedback to any earlier stage: re-collecting or relabelling data, engineering new representations, retraining, or even redefining the business problem. Real environments change, so a one-way process would lock in early mistakes. The figure shows "continuous feedback drives improvement" from monitoring back to problem formulation.

Q58. What elements should a robust AI problem statement define before model development begins?

The target population and the decision-maker; the specific decision and the exact time it occurs; the desired outcome and available alternatives (including baseline/status quo); the consequences of errors (FP vs FN); and the operational and ethical constraints. In worksheet form: decision time, population/unit, available information, output, horizon, decision-maker, action, baseline, metrics, error costs, subgroup risks, constraints and governance. "If this statement cannot be completed clearly, model development should not begin."

Q59. Why is the statement "build a model to predict hospital readmission" an inadequate problem formulation?

It is "vague, lacks operational context, and ignores the downstream impact". It does not say when the prediction is made (admission? discharge?), which patients are included, what counts as readmission (planned or unplanned) or within what window, who uses the output, what action follows, what resource is limited,

what the baseline is, or which risks (calibration, subgroup fairness, false alerts) must be checked. Without these, features could leak, the model could be accurate but useless, and no one could judge success.

Q60. What additional details make a hospital readmission problem operationally meaningful?

From the better framing: **when** (at the time of discharge); **who uses it** (the discharge team); **what action** (allocating follow-up appointments); **what resource is constrained** (a limited number of appointment slots); **which patients** (adult medical patients); **what outcome period** (unplanned 30-day readmission); **what risks are examined** (calibration, subgroup fairness, preventability and the consequences of false alerts).

Q61. What questions should be asked about data provenance, variable definitions, completeness, temporal integrity, and governance?

- **Provenance and purpose:** Who or what produced the data, and for what original purpose?
- **Definitions and proxies:** Under which definitions were variables established? Which are direct measurements and which are proxies?
- **Completeness and consistency:** Which cases are missing? Was the outcome recorded consistently?
- **Temporal and relational integrity:** What has changed over time? Are timestamps reliable? Can records be linked correctly?
- **Governance:** Are the data legally and ethically permissible for this specific objective?

Q62. What is the purpose of a Datasheet for Datasets?

Proposed by Gebru et al. and inspired by electronics datasheets, it documents a dataset's **motivation, composition, collection methods, preprocessing steps, recommended uses and limitations**. It makes implicit assumptions, biases and constraints explicit, so developers and downstream users understand the data's boundaries, reduce misuse, and build on "a foundation of transparency and informed consent".

Q63. How can poor dataset documentation lead to unsafe deployment of a skin lesion classifier?

The 100,000-image dataset was about 90% fair-skinned patients, mostly from specialized clinics in one country, captured with high-end clinical cameras (while many hospitals use smartphones), and labelled by single dermatologists without verification. None of this was documented, so developers assumed it was representative and hospitals adopted it. Months later it performed much worse for darker skin. A datasheet listing these limitations ("underrepresentation of darker skin tones, children, and smartphone images; not validated for primary care") would have prompted more data collection, fairness evaluation, or restricted deployment.

Q64. Why is representation described as the bridge between the physical world and computational models?

Algorithms cannot process people, images, documents or road networks directly; representation translates them into structured mathematical forms: a person into a feature vector, an image into a tensor of pixel intensities, text into tokens or embeddings, a road network into a graph, purchases into an event sequence, a physical process into a state-space model, categories into one-hot or learned embeddings. Representation "dictates exactly which information is visible to the model", so it determines which patterns are easy or hard to learn.

Q65. Explain loss of context, proxy encoding, aggregation bias, temporal leakage, identity errors, and train-serving mismatch.

See Part 2, section 13 for the table. Short versions with the ICU sepsis example:

- **Loss of context:** a 24-hour average heart rate hides a dangerous spike one hour ago.
- **Proxy encoding:** home zip code encodes socioeconomic status and race, so alerts are deprioritized for marginalized neighbourhoods.
- **Aggregation bias:** 92% overall accuracy hides failure for paediatric and autoimmune patients.
- **Temporal leakage:** antibiotic dose counts updated after the diagnosis let the model "wait for the doctor's order" (99% test accuracy, fails live).
- **Identity errors:** a transferred patient split into two IDs, hiding the deterioration trend.
- **Train-serving mismatch:** missing BP imputed with the ward median in training but with 0 in production, flooding nurses with false alarms and causing alert fatigue.

Q66. What does the mapping f_θ: X → Y represent in model development?

A model is a mathematical function that maps structured input representations X (the feature space) to outputs Y (predictions or decisions), with parameters θ learned from data. It can be a linear function, decision tree, nearest-neighbour procedure, probabilistic graphical model, deep network, search or optimization program, rule set, or a neuro-symbolic hybrid. Optimization is not guaranteed to find the global optimum; it may reach a local, approximate or regularized solution.

Q67. Which operational, governance, data, and lifecycle criteria should guide model selection?

- **Data characteristics:** size, dimensionality, nonlinearity.
- **Task requirements:** classification, regression, ranking; need for probabilistic calibration.
- **Operational constraints:** latency, memory, computational limits for real-time inference.
- **Governance and trust:** interpretability, explainability, robustness required by stakeholders or regulators.
- **Lifecycle costs:** long-term maintenance burden and ML expertise available in the team.

"The choice should never be based on hype."

Q68. Why can a simpler model be preferable to a more accurate but complex model?

Loan example: DNN 94% accuracy vs logistic regression 91% with 15 features. The simpler model wins because: **interpretability** (ECOA requires specific adverse-action reasons; logistic coefficients give them directly, while post-hoc SHAP on a DNN can be unstable); **data size** (50,000 applications a year is too small for a massive DNN without overfitting); **maintenance cost** (after economic shifts, standard risk analysts can maintain logistic regression, while a DNN needs specialized ML engineers); **latency and compute** (logistic regression runs on cheap CPUs). A slightly less accurate model that fits the constraints delivers more real-world value.

Q69. Why should the principle of parsimony be treated as a heuristic rather than as a rule that reality must always be simple?

Occam's principle says prefer the simplest model only when competing explanations fit the evidence equally well. It does not claim reality is simple. A physician who insists that tuberculosis alone explains cough, fever, weight loss and joint pain misses coexisting rheumatoid arthritis and delays treatment. Because the two-disease model fits the evidence better, it is the right choice despite being more complex. Parsimony should "encourage avoiding unnecessary complexity, not ignoring necessary complexity."

Q70. What information should a Model Card communicate to downstream users?

Following the notes' heart-disease Model Card: **Model Overview** (name, purpose, type, architecture, intended use); **Data Description** (data source, features, preprocessing such as mean/mode imputation, exclusions such

as race or genetics); **Performance Metrics** (overall accuracy, precision, recall, F1, performance by subgroup, fairness observations); **Known Limitations** (bias in data, generalizability, feature limitations); **Intended and Inappropriate Uses**; **Ethical Considerations** (fairness, impact of misuse); **Additional Considerations** (future work). Section 1.5.5 summarizes the purpose: record the intended use, evaluation procedures, conditions under which it performs well, detailed subgroup results and known limitations, giving users "a clear, honest picture of the system's capabilities and boundaries". (A complete example is in A2, Part 8.)

Evaluation, Decisions, Deployment, and Monitoring

Q71. Why is overall accuracy insufficient for evaluating a high-stakes AI system?

Accuracy treats all errors as equally costly and can hide: poor calibration (80% predicted risk but 45% observed deterioration in the ED triage example); subgroup failures (lower accuracy for elderly and non-English speakers); fragility (crashing when one vital sign is missing); operational problems (15-second latency, worse on night shifts); and lack of real-world benefit (only 2% better than the existing early warning score, and alert fatigue destroyed decision utility). It is also misleading with class imbalance. The real question is whether the system is "sufficiently reliable, safe, and effective for its specific intended use".

Q72. Distinguish discrimination from calibration.

- **Discrimination:** the model's ability to separate or rank positive cases above negative ones (measured by AUROC).
- **Calibration:** whether predicted probabilities match observed frequencies; a predicted 80% risk should correspond to about 80% observed events.

A model can discriminate well yet be badly calibrated (readmission example: ranks well, but 80% predicted means 35% observed). Calibration matters whenever the probability itself is used for decisions or resource allocation.

Q73. What evaluation dimensions should be assessed in addition to predictive performance?

- **Fairness and robustness:** subgroup performance, robustness to noise and distribution shift, interpretability.
- **Operational viability:** computational cost, inference latency, stability across times, environments and sites.
- **Real-world impact:** decision utility, comparison with meaningful baselines, and human-system performance (how humans and AI perform together).

Q74. Why must temporal, geographic, equity, probabilistic, and decision claims be matched with appropriate validation methods?

Because a test only supports the claim it actually checks. Readmission example:

- **Temporal claim:** random 80/20 split gave 85%; training on 2022 and testing on January 2023 gave 68%, exposing a billing code used only for three months in 2021.
- **Geographic claim:** strong at the urban flagship; external validation at rural clinics failed because a lab test only existed at the flagship.
- **Equity claim:** AUC 0.82 overall but 0.55 for non-English speakers.
- **Probabilistic claim:** calibration curve showed 80% predicted vs 35% observed.
- **Decision claim:** a prospective pilot showed 60% flagged, alert fatigue, and nurses reverting to spreadsheets.

"How you test an AI is just as important as what you test."

Q75. Why is a machine learning prediction not the same as a decision?

A model outputs a probability $\hat{p} = \hat{P}(Y = 1 \mid X = x)$; a decision requires a concrete action. Converting one to the other requires a decision rule (intervene if $\hat{p} \geq t$) whose threshold reflects error costs, intervention capacity, uncertainty, fairness and human review. The same prediction can lead to different actions under different capacities or values. "A prediction is just a number; the decision is where human values and operational constraints are actually applied."

Q76. How should asymmetric error costs, intervention capacity, uncertainty, fairness, and human review influence a decision threshold?

Rapid Response Team example:

- **Asymmetric costs:** a missed deterioration (FN) can mean a preventable death while a false activation (FP) costs a brief assessment, so t is pushed below 0.5.
- **Intervention capacity:** only two RRT nurses at night; t = 0.2 might flag 40 patients, so t is raised enough to stay within capacity.
- **Uncertainty and expected benefit:** for borderline $\hat{p}$ (0.35 to 0.45) trigger a lower-acuity action (more frequent vital-sign checks) instead of a full activation.
- **Fairness:** a single global t gave a high FN rate for non-English speakers; recalibrate, improve data, retrain, or where appropriate adjust thresholds.
- **Human-in-the-loop:** a two-minute nurse review before dispatch allows a lower t (0.3) if false positives stay manageable.

The threshold is a "tunable policy lever", not a mathematical default.

Q77. Interpret the conceptual decision-cost function that combines false positives, false negatives, delay, and workflow cost.

$$C(t) = c_FP \cdot P(FP|t) + c_FN \cdot P(FN|t) + c_delay \cdot E[D|t] + c_workflow(t)$$

The system chooses the threshold t that minimizes total real-world cost: the cost of false alarms times their probability, the cost of misses times their probability, the cost of expected delay or friction, and the cost of fitting the threshold into the human workflow. Lowering t reduces misses but increases false alarms, queues and workload. Baggage example: t = 0.4 maximized accuracy (96%) but created huge manual-search queues and broke the two-station checkpoint; t = 0.7 (94% accuracy) minimized C(t). "The optimal decision threshold is rarely the one that simply maximizes overall accuracy."

Q78. What engineering and governance questions must be answered before moving a model from a sandbox into production?

- **Infrastructure and performance:** Where will inference occur (cloud, edge, on-premise)? What is the maximum latency? Which features are actually available in real time?
- **Resilience and failure handling:** What happens when inputs are missing, delayed or corrupted? Can the system abstain? What is the fallback? Can it be rolled back instantly?
- **Human-system interaction:** Who receives the output? How are reasoning and uncertainty explained?
- **Governance and MLOps:** How are model versions, code and pipelines controlled and tracked? Who is accountable for monitoring, retraining and maintenance?

Q79. What is the role of abstention, fallback procedures, rollback mechanisms, and circuit breakers in deployment?

They make the system fail safely. **Abstention:** the model declines to predict when confidence is too low and defers to a human or another process. **Fallback:** a defined safe path when inputs are missing or errors occur (in the fraud example, a missing geolocation feature routes the transaction to the legacy rule-based system). **Rollback:** reverting instantly to a previous stable version. **Circuit breaker:** an automatic trigger for rollback; if the decline rate spikes more than 5% in 10 minutes, the system rolls back to protect legitimate customers.

Q80. What should be monitored across input quality, drift, feature pipelines, model outputs, predictive performance, operations, human interaction, fairness, and feedback loops?

- **Input quality:** missing values, invalid ranges, unexpected formats, schema changes (an app update alters telemetry).
- **Input distribution (data drift):** shifts in demographics, devices, regions, language, case mix.
- **Concept and label drift:** changing relation between inputs and outcomes (fraudsters change tactics).
- **Feature pipelines:** transformation errors, delayed joins, missing updates, stale features.
- **Model outputs:** prediction distributions, confidence, uncertainty, abstention rates.
- **Predictive performance and ground truth:** error rates, calibration, precision, recall, F1, subgroup metrics; availability and timeliness of labels.
- **Operational health:** latency, uptime, throughput, CPU/GPU/memory.
- **Human-system interaction:** override rates, acceptance, corrections, automation bias.
- **Harm, fairness and safety:** unequal error rates, harmful outcomes.
- **Feedback loops:** predictions shaping future training data.

Types of AI Problems

Q81. Distinguish binary, multiclass, and multilabel classification.

- **Binary:** $Y \in \{0, 1\}$; fraud vs legitimate, disease vs no disease, spam vs non-spam.
- **Multiclass:** $Y \in \{1, \ldots, K\}$; exactly one class is chosen (handwritten digit).
- **Multilabel:** $Y \subseteq \{1, \ldots, K\}$; several labels may apply at once (several diagnoses on one clinical record).

Q82. Why must a classification label such as "high risk" include an event definition, threshold, and time horizon?

"High risk" alone is not operationally meaningful: high risk of what event, how high, and by when? Without an event definition the label cannot be measured consistently; without a threshold nobody knows which scores count as high; without a horizon the prediction cannot be matched to an intervention window. Example: "unplanned readmission within 30 days, $\hat{p} \geq t$" (or in RYDA, not in verified employment at day 180 and ranked in the top 300).

Q83. Compare mean squared error and mean absolute error for regression tasks.

$$\text{L_MSE}(y, \hat{y}) = (y - \hat{y})^2 \quad \text{L_MAE}(y, \hat{y}) = |y - \hat{y}|$$

MSE squares errors, so it penalizes large errors heavily (useful when big misses are dangerous). MAE grows linearly, so it is more robust to large deviations and outliers. The choice "encodes an assumption about error".

Q84. Why should the regression loss function reflect the real application rather than convention alone?

The loss defines what the model treats as a bad mistake. If real harm grows sharply with error size (under-predicting energy demand causing blackouts), MSE fits; if outliers are noise, MAE fits; if over- and under-prediction have different costs (asymmetric cost), neither symmetric loss is right. Table 2 lists ignoring skew, heteroscedasticity or asymmetric cost as the common regression framing error. Choosing by habit optimizes the wrong thing (optimization is blind to whether "the loss function reflects real harm").

Q85. How do ranking and recommendation differ from ordinary classification?

Ranking produces an ordered list and is evaluated by list-aware metrics (NDCG, MAP, precision@k); items cannot be judged independently of their position. Recommendation adds personalization and availability constraints (recall@k, NDCG, utility, diversity). Crucially, a recommender "selects what the user will be shown", influencing future preferences and creating the feedback loop: past exposure → observed interaction → training data → future exposure. Ordinary classification labels items independently and does not choose what data it will see next.

Q86. Why are observed clicks not pure measures of user preference in a recommendation system?

Clicks are "conditional on previous rankings, interface design, position, and availability". Users can only click what they are shown, and items shown near the top get more clicks regardless of quality (position bias). Those clicks then become training data, so the system learns its own past choices (selective exposure, feedback loop). Optimizing clicks can also differ from long-term benefit.

Q87. Why should clusters not automatically be treated as objectively real categories?

Clustering finds structure without labels, and the result depends on the representation, scaling, algorithm and number of clusters. "A mathematically compact partition may still be scientifically meaningless." So clustering requires stability analysis, sensitivity to representation and scaling, inspection of cluster composition, domain validation, and caution against reifying algorithmic groups as natural categories (for example, customer segments are useful summaries, not natural kinds).

Q88. What components are required to formulate a search problem?

An initial state, actions or transitions, a goal test, a path cost, and possibly a heuristic (a problem-specific estimate that guides search toward the goal, such as straight-line distance in A*). The common framing error is defining an incomplete search space.

Q89. Compare search, planning, and control as classical AI problem types.

- **Search** explores a space of states or candidates to find a goal state, path or document (route finding, document retrieval); evaluated by relevance, path cost, success rate, latency.
- **Planning** identifies a sequence of actions expected to achieve a goal while satisfying constraints (robot task planning); error: assuming a static or fully known environment.
- **Control** repeatedly observes a system and acts over time: x_{t+1} = F(x_t, u_t, w_t), with state x, action u and disturbance w (keeping a drone on its trajectory); evaluated by stability, tracking error, safety, energy. "Unlike static prediction, control changes the process being predicted."

Q90. Explain the main framing risks in anomaly detection, generation, forecasting, causal estimation, and decision optimization.

- **Anomaly detection:** "unusual is not the same as undesirable"; anomalies may be fraud or legitimate novelty; without an investigation or response policy it only produces unusualness scores (error: assuming all rare cases are harmful).
- **Generation:** a fluent output can still be false, unsafe, biased or irrelevant; evaluate plausibility, grounded correctness, and utility and safety (error: equating fluency with truth).
- **Forecasting:** time order matters; random splits leak the future; use temporal holdouts or rolling-origin validation, compare with naive or seasonal baselines, and watch for trend, seasonality, autocorrelation and structural change.
- **Causal estimation:** $\tau = E[Y(1) - Y(0)]$ but both outcomes are never seen for the same unit; needs randomization, exchangeability, positivity, consistency, correct temporal ordering and control of confounding (error: interpreting correlation as intervention effect).
- **Decision optimization:** $a^* = \text{argmax } E[U(a, Y) \mid X = x]$; errors include optimizing an incomplete proxy objective, treating risk as intervention benefit, ignoring constraints, and evaluating predictions without evaluating the resulting actions.

Problem Formulation and Common Failure Modes

Q91. Why should an AI project begin with a decision or capability rather than with an available dataset?

The data-first trap (grab a convenient dataset, pick an interesting target, maximize AUC, then look for a use case) produces "highly accurate mathematical curiosities that deliver zero real-world value". Attrition example: an 88%-AUC model listed 500 at-risk employees, but HR had no budget or time to act and the model was abandoned. Starting from the decision (which employees should managers target, with only five interventions per quarter) led to a transparent heuristic that caught 80% of churners and was trusted and used. "A model that does not change a decision or trigger a concrete action has zero operational value."

Q92. What questions are included in a comprehensive problem-framing canvas?

- **The Decision:** What specific decision will be altered, informed or automated?
- **The Actor:** Who is the decision-maker responsible for executing it?
- **The Actions:** What concrete interventions are available once the output is generated?
- **The Necessity of AI:** Is a complex ML model actually required?
- **The Simpler Alternatives:** Could a deterministic rule, checklist, statistical summary or process redesign do it more cheaply or reliably?

Q93. Define the target population, unit of analysis, index time, prediction horizon, inputs, outputs, labels, constraints, stakeholders, and cost of error.

- **Target population:** the set of cases the system is meant for, with inclusion and exclusion criteria (adult medical patients).
- **Unit of analysis:** the independent entity being predicted for: individual, event, object or spatial entity.
- **Index time:** the exact moment the prediction is made.
- **Prediction horizon:** the future window in which the outcome must occur.
- **Inputs:** documented features (definition and source, unit, update frequency, missingness, timestamp semantics, availability at inference, privacy class, transformation, known biases).

- **Outputs:** class, value, probability, interval, ranking, score, tier, recommendation, action, artifact, or abstention.
- **Labels:** the ground truth used for training, often a proxy that must be interrogated.
- **Constraints:** technical, resource, trust and usability, governance and legal boundaries.
- **Stakeholders:** subjects, targets, operators and experts, and the broader ecosystem.
- **Cost of error:** the real consequences of TP, FN, FP, TN, which are usually asymmetric and include non-financial harms.

Q94. Why must every feature be available at or before the index time?

Because the deployed model can only use what is known at the moment it predicts. A feature recorded later carries future information; it looks very predictive in retrospective data but is unavailable (or different) in real time. "Using that result does not constitute early prediction; it is merely hindsight." This is target/temporal leakage: "If a model relies on data that has not yet occurred at the index time, it is leaking the future, not demonstrating intelligence." Data engineering must also check availability, not just event time: t_feature availability ≤ t.

Q95. How should a project team define a meaningful baseline and an evaluation plan before training begins?

Baseline: the real alternative the system will replace or augment, not random guessing: current professional judgment, a simple heuristic, the status quo, a legacy system, or alternative policies (for example random allocation). The question becomes "Does deploying this AI system improve real-world outcomes compared to the best available alternative?"

Evaluation plan (locked before training): primary and secondary (guardrail) metrics; validation strategy (external, temporal and prospective testing); reliability and fairness (subgroup analysis, calibration, uncertainty, robustness); operationalization and governance (decision thresholds, acceptance criteria, monitoring plan). Locking it in prevents metric shopping and data dredging: "An evaluation plan written after the fact is merely a retrospective justification for deployment."

Q96. Explain how solving the wrong problem, poor data quality, data leakage, weak evaluation, underspecification, distribution shift, ignored workflow, and absent monitoring can cause AI failure.

- **Wrong problem:** the target is disconnected from the goal (predicting cost instead of preventable illness; clicks instead of welfare; risk with no capacity to act).
- **Poor data quality:** wrong values, missing records, inconsistent units, duplicate identities, delayed updates, coding changes, unrepresentative samples, unreliable labels, undocumented preprocessing; more data does not fix systematic bias.
- **Data leakage:** target, temporal, group, preprocessing, label leakage and benchmark contamination give unusually impressive but fake results.
- **Weak evaluation:** only accuracy, convenient test sets, ignored imbalance, test-set tuning, no CIs, no baseline, no calibration, no subgroups, random splits for temporal use.
- **Underspecification:** several models with equal test scores rely on different shortcuts and behave differently under shift; mitigate with stress tests, multiple runs, shift and subgroup evaluation, causal checks.
- **Distribution shift:** populations, devices, policies, seasons, users, economy or pipelines change.
- **Ignored workflow:** alerts too late, distrust, poor explanations, too many alerts, automation bias, unclear responsibility, infeasible actions, more workload.

- **No monitoring:** silent degradation when schemas change, features fail, labels are delayed, thresholds change, new subpopulations appear, or versions are not recorded.

Data Engineering and Illustrative Case Studies

Q97. Why is data engineering described as the connective tissue of the complete AI lifecycle?

Every stage depends on data flowing reliably: generated, captured, validated, stored, transformed, served, observed and governed. Every model needs data with defined meaning, delivered consistently to training and production; labels and outcomes must be linked over time; behaviour must be monitored; decisions audited; privacy enforced; source changes detected; failures made reproducible and reversible. So data engineering is "not merely what happens before machine learning"; it connects problem formulation to monitoring. "Reliable model ⇏ Reliable AI system."

Q98. How do semantics, temporal correctness, training-serving consistency, lineage, governance, and monitoring make data engineering unavoidable?

- **Semantics:** "temperature" needs units, site, sensor, timestamp, missing code and calibration status; "diagnosis_date" could mean symptom onset, documentation, confirmation, entry or billing date.
- **Temporal correctness:** features must be recorded and available by prediction time (t_feature availability ≤ t); no train/test overlap; no stale data treated as real time.
- **Training-serving consistency:** encodings, missing-value handling, units, windows, library versions and batch vs streaming must match; use shared versioned transformations and tests.
- **Lineage:** source version, query, transformation code, label definition, partitions, seed, libraries, configuration, evaluation data, approval and deployment version, for audit and rollback.
- **Governance:** role-based access, encryption, retention limits, de-identification, consent tracking, audit logs, purpose restrictions, deletion workflows, incident response.
- **Monitoring:** only possible if inputs, predictions, versions, human responses, outcomes and subgroup data are captured.

Q99. What lessons about proxies, drift, lineage, and ground-truth anchoring can be learned from Google Flu Trends and the Epic Sepsis Model?

Google Flu Trends: search behaviour ≈ disease activity is only a proxy, affected by media, anxiety, platform design and habits. It overestimated flu in 100 of 108 weeks and exceeded CDC estimates by more than 2× in February 2013. Causes: big-data hubris (about 50 million candidate terms against 1,152 weekly observations, chance correlations) and algorithm dynamics (the platform changed the data-generating process). Lessons: volume does not remove measurement bias; a proxy is useful only while its relationship to the target is stable; platforms are part of the data-generating process; undocumented term lists destroy reproducibility (lineage); combining GFT with lagged CDC data cut MAE to 0.232 (vs 0.486 and 0.311), so new sources should complement ground truth; monitor residual drift and suspend when it grows.

Epic Sepsis Model: external validation (38,455 hospitalizations) found AUROC 0.63, alerts on 18% of hospitalizations, 67% of sepsis cases missed at threshold ≥ 6, and only 183 cases caught before timely antibiotics. Lessons: the label must match the intended clinical event (labels built from clinician actions encode practice); timing is as important as accuracy (predicting after antibiotics are ordered is useless); features need event, recording, availability and correction times; missingness encodes clinician behaviour; sites differ, so external and prospective validation are essential; alert volume is a system-level metric; workflow decides value.

> **Q100. How does TensorFlow Extended demonstrate that production AI depends on ingestion, statistics, schemas, validation, transformation, training, evaluation, model validation, serving, monitoring, and lineage?**

TFX turns a one-off training script into a repeatable pipeline: **ingestion** with versioned specs, record counts and checksums; **statistics** on distributions, missingness, ranges and label balance; a **schema** that makes assumptions machine-checkable; **data validation** catching missing columns, new categories and shifts; **transformations** defined once for training and serving (no skew); **training** that records code, data, hyperparameters and seeds; **evaluation** against the current production model; **model validation** that blocks release if metrics, subgroup performance, fairness or latency fail; **serving and monitoring** with version metadata and rollback. Remove any piece and reliability drops (silent corruption, skew, irreproducible results, worse models shipped, invisible failures). Business value (months to weeks deployment; about 2% more Google Play installs) came from the whole system. "Data engineering ... constitutes much of the production architecture."

Part 6: Scenario Questions S1 to S5

These are the most "quiz-like" questions: a short case, then diagnosis questions. Practise answering each bullet in 3 to 6 lines using the named concept.

S1: University Student Support System

> **S1(a). Why is the original problem statement ("Build a machine learning model that predicts student failure") inadequately formulated?**

It is the "poor framing" pattern from the notes. It starts from an algorithm instead of a decision and leaves every key element undefined:

- **No index time:** we do not know when the prediction is made, so we cannot tell which features are legitimate (leakage risk).
- **No target population:** which course, which students (all enrolled, first-years, repeaters)? Who is excluded (students who withdrew before the census date)?
- **No outcome definition:** does "failure" mean a final grade below the pass mark, withdrawal, or both? Over what period?
- **No link to the action or capacity:** the purpose is to allocate a limited number of tutoring appointments, but the number is not stated, so we cannot design a ranking or threshold.
- **No decision-maker, baseline, metrics, error costs or subgroup analysis.** This risks "Failure 1: predicting student failure without an effective support intervention."

> **S1(b). Rewrite the problem statement by specifying the target population, index time, prediction horizon, outcome, decision-maker, and intended action.**

"At the end of week 4 of the semester, for every student enrolled in [Course X] on the census date (one student-course enrolment is the unit of analysis), using only information recorded by the end of week 4 (prior academic performance, attendance and assignment grades from weeks 1 to 4, learning-platform activity in weeks 1 to 4, and financial-aid status as recorded at enrolment), produce a calibrated probability that the student will fail the course (final grade below the pass mark, or withdraw after the census date) by the end of the semester, together with a ranked list, so that the course coordinator and academic-support office can offer the N available tutoring appointments to the highest-ranked students after human review. The system will be compared with the current process (self-referral and instructor referral) and evaluated using recall@N, precision@N, calibration and subgroup performance, accounting for the cost of missed struggling students, stigma from being flagged, and fairness across financial-aid and demographic groups."

> **S1(c). Which variables may create temporal leakage if the prediction is required during the first four weeks of the semester?**

- **Final examination results:** these happen at the end of the semester and are part of the outcome itself (target leakage).
- **Assignment grades after week 4** and cumulative course grades that include later work.
- **Attendance and learning-platform activity after week 4** (future behaviour; declining activity late in the term reflects the outcome developing).
- **Midterm results** if the midterm occurs after week 4.
- **Financial-aid status if it was updated later** (for example aid suspended because of poor grades); only the value recorded at the index time can be used. Also any withdrawal records or academic-probation flags.

Rule: every feature must be recorded and available by the end of week 4 ($t_{feature\ availability} \leq t$).

S1(d). What are the likely consequences of false-positive and false-negative predictions?

- **False positive** (flagged but would have passed): a scarce tutoring slot is spent on a student who did not need it, displacing someone who did; possible stigma, anxiety or feeling labelled; wasted tutor time.
- **False negative** (not flagged but fails): the student misses support when it could have helped, fails or withdraws, loses time, money, scholarship eligibility, and may drop out. This is usually the more costly and less reversible error, so recall on at-risk students is important, bounded by the appointment capacity.

S1(e). Should the university evaluate only predictive accuracy, or also calibration, subgroup performance, and decision utility?

It should evaluate all of them. Accuracy is misleading when most students pass (class imbalance) and treats all errors equally. **Calibration** matters because probabilities are used to rank students and decide who gets scarce appointments. **Subgroup performance** matters because financial-aid status and platform activity may act as proxies for socioeconomic status, and errors may concentrate in some groups. **Decision utility** matters because the real goal is fewer failures: the system must be compared with the current referral process and ideally tested (for example a pilot) to see whether tutoring the flagged students actually improves outcomes.

S2: Hospital Sepsis Prediction

S2(a). Identify the train-serving mismatch in this case.

In training, missing blood-pressure values were imputed with the median for the relevant ward. In production, the software team imputes missing values with zero. A blood pressure of 0 is physiologically impossible and looks like extreme shock to the model, so it produces floods of high-priority false alarms. This is exactly the notes' **train-serving mismatch**: different preprocessing pipelines in training and live inference, causing "silent and often catastrophic performance degradation" and alert fatigue.

S2(b). Explain why the antibiotic-dose feature may introduce temporal leakage.

The number of broad-spectrum antibiotic doses is often updated after a physician has already suspected or diagnosed sepsis. So in historical records the feature carries information about the diagnosis itself. The model learns to "wait for the doctor's order": it detects sepsis by detecting the treatment. At the real index time (six hours before sepsis), those doses have not yet been given, so the feature is unavailable or systematically different. It is future information leaking into the past (and a form of target leakage because the feature is a consequence of the outcome).

S2(c). Why can high offline accuracy coexist with poor real-world performance?

- The leaked antibiotic feature inflates offline accuracy but is absent at prediction time.
- The imputation mismatch changes inputs in production.
- Accuracy is dominated by the many non-sepsis patients (low prevalence), so 98% accuracy can hide poor detection of actual sepsis and a low positive predictive value.
- Offline evaluation may have used random splits and not tested temporal, external or workflow claims. The Epic Sepsis Model shows the same pattern: AUROC 0.63 externally, alerts on 18% of hospitalizations, 67% of cases missed.

S2(d). Which data validation checks should be placed between data ingestion and model inference?

- **Schema checks:** required fields present, correct data types and units (mmHg).

- **Range checks:** physiologically plausible values (reject BP = 0, heart rate > 300).
- **Missingness checks:** rate of missing values per feature compared with the reference; flag spikes.
- **Imputation consistency test:** the same versioned transformation code used in training and serving (feature store); automated test comparing training and serving outputs for the same record.
- **Timestamp checks:** event time, recording time, availability time; reject features whose availability time is after the index time.
- **Distribution/drift checks:** compare incoming distributions with training statistics (TFX-style statistics + schema + validation).
- **Identity checks:** patient and encounter IDs resolve correctly (no split identities after transfers).

S2(e). What fallback action should occur when critical clinical inputs are missing or invalid?

The model should **abstain** rather than impute a misleading value: no risk score is produced (or it is clearly marked "insufficient data"), and the case is routed to the standard clinical process, such as the existing early-warning score or nurse assessment, with a prompt to re-measure the vital sign. Missing-input events should be logged and monitored; if the rate spikes, a circuit breaker should suspend alerts and fall back to the legacy protocol while the pipeline is fixed.

S3: Automated Loan Underwriting

S3(a). Which model should the bank consider for deployment, and why?

The **logistic regression** model (91%). This is the notes' Example 25. The bank must give applicants clear reasons for rejection (adverse-action reasons, as under the Equal Credit Opportunity Act); logistic regression gives directly interpretable coefficients, while the DNN's explanations vary across post-hoc methods (unstable SHAP-type explanations). The bank also has limited ML engineering capacity, and logistic regression uses 15 established financial variables on standard hardware, making it easier to maintain, audit, recalibrate and monitor. The 3-point accuracy gain does not outweigh legal, governance and lifecycle costs.

S3(b). Why is the model with the highest predictive accuracy not automatically the best operational choice?

Because accuracy is only one criterion. "The most complex model is not automatically the best model." A model must also meet legal explainability requirements, run within compute and latency budgets, be maintainable by the available team, be robust to drift (thousands of behavioural features may shift or act as proxies), and be auditable. A model that cannot be explained or maintained will be unusable or risky, however accurate offline.

S3(c). Which criteria beyond accuracy should guide model selection?

Data characteristics (size relative to model complexity; about 50,000 applications a year is small for a large DNN); task requirements (calibrated default probabilities); operational constraints (latency, memory, compute); governance and trust (interpretability, explainability, robustness, fairness across groups, proxy risk in behavioural features); lifecycle costs (maintenance burden, team expertise, retraining frequency); plus calibration, subgroup performance and external/temporal validation.

S3(d). How do interpretability, latency, maintenance cost, and governance requirements affect the decision?

- **Interpretability:** decisive. Regulations require specific reasons; coefficients provide stable, auditable reasons, while variable post-hoc explanations do not.

- **Latency:** both may respond in milliseconds, but the DNN needs much more computing resources (CPUs/GPUs) to keep that speed at scale; logistic regression runs cheaply on standard CPUs.
- **Maintenance cost:** when macroeconomic conditions shift (inflation spike), relationships change; a DNN needs specialized ML engineers to monitor, recalibrate and retrain, whereas standard risk analysts can maintain logistic regression.
- **Governance:** auditors and regulators need to understand and challenge decisions; thousands of behavioural features raise proxy-discrimination and privacy risks that are hard to audit.

S3(e). What information should be documented in a Model Card before deployment?

Model overview (name, version, type: logistic regression binary classifier of default within a defined horizon; purpose; intended use as decision support for underwriters); data description (source and period of loan records, the 15 features and definitions, preprocessing and missing-value handling, exclusions); performance (accuracy, precision, recall, AUROC, calibration curve, confidence intervals, temporal validation results); subgroup performance by protected and proxy-sensitive groups (age, sex, region, thin-credit-file applicants); fairness observations; known limitations (limited credit history, economic regime changes); intended and inappropriate uses (not for automated rejection without review, not for other products or countries without validation); the decision threshold and its justification; adverse-action reason generation; monitoring plan and triggers for recalibration or rollback; and the accountable owner.

S4: AI-Based Recruitment Ranking

S4(a). Why should this system be understood as a socio-technical system rather than only as a ranking algorithm?

Its impact depends on all five layers: the **model** (ranks applicants), the **data infrastructure** (ten years of past hiring decisions), the **interface** (top ten in green, the rest grey), the **human workflow** (recruiters told to review all, but invitations go mostly to the highlighted ten), and the **institutional rules** (retraining on new hires, employment law, fairness policy). The outcome, who gets interviewed, is produced by this whole composite, not by the ranking function alone. This is the notes' Example 12.

S4(b). How may historical hiring data encode institutional bias?

The label is "who was hired in the past", which records past recruiter decisions, not true candidate quality (status quo bias). If recruiters favoured a small group of universities (because of networks, prestige or familiarity), the model learns university as a strong signal. University then acts as a proxy for socioeconomic status, geography and other protected attributes (proxy encoding). The model reproduces the old preference "faster and at a much larger scale".

S4(c). How can the dashboard design influence human decisions and create automation bias?

Colour-highlighting the top ten signals "these are the right ones" and makes accepting the ranking the easiest option; the grey candidates look pre-rejected. Automation bias is "the tendency of humans to place excessive trust in automated systems and accept their outputs without sufficient independent evaluation". Although recruiters are instructed to review everyone, most invitations go to the green candidates, so the formal instruction is overridden by the interface. There is no uncertainty information, so small score differences look meaningful.

S4(d). Explain how retraining on the new hiring data may create a self-reinforcing feedback loop.

Candidates from the favoured universities are ranked high → highlighted → interviewed and hired → recorded as successful. Candidates from other universities are rarely interviewed, so their potential success is never

observed (selective labels). Retraining on these new hires therefore shows an even stronger link between the favoured universities and hiring. The model's output has become its input: past exposure → observed interaction → training data → future exposure. The bias grows and the model looks increasingly "accurate" because it is judged on people it selected.

S4(e). Which stakeholders should be responsible for governing, measuring, and managing the risks of this system?

Apply NIST: **Govern:** senior HR leadership and the organization's AI governance/ethics committee, with legal/compliance, set policy and name an accountable owner who can suspend the system. **Map:** HR, legal, diversity and inclusion teams and data scientists identify risks (proxy discrimination, automation bias, feedback loops) and affected groups (applicants from non-favoured universities). **Measure:** data scientists and independent auditors track selection rates and error rates by subgroup, override rates, drift and feedback effects. **Manage:** product/interface designers remove the green/grey framing and add uncertainty; recruiters and hiring managers keep decision authority and log overrides; ML engineers use counterfactual logging and stop naive retraining; an appeal channel for applicants. Applicants and regulators are stakeholders who must be informed and consulted.

S5: Smart Warehouse Robot

S5(a). Why is this system an example of hybrid AI?

It combines several computational approaches, each used where it fits: deep learning for noisy visual perception, search for routing, optimization for travel time, explicit rules for restricted areas, a safety controller for collisions, and human review for damaged packages. Neither a purely symbolic system (cannot interpret noisy camera images) nor a purely statistical system (might violate safety rules in unfamiliar situations) would be sufficient. This is the notes' Example 9.

S5(b). Which components are statistical, symbolic, search-based, optimization-based, and human-controlled?

Category	Component
Statistical (learned)	Deep learning model for package and obstacle recognition
Symbolic (rule-based)	Explicit rules prohibiting entry into restricted areas; the safety controller's stop rule (collision-risk threshold)
Search-based	Search algorithm for route selection
Optimization-based	Procedure that minimizes travel time
Human-controlled	Review and final decision on damaged packages

S5(c). What types of data drift or environmental change are present?

- **Environment/layout change:** the warehouse layout changes, so the map and learned traversable paths are outdated (the search space and costs changed); routes get longer.
- **Input distribution drift (new device/appearance):** new reflective packaging materials look different from training images, so the perception model fails to recognize them (covariate shift and possibly a new unseen class).
- **Population/context change:** more workers in the facility (more people and obstacles, more congestion), which changes traffic patterns and the frequency of safety stops.
- **Seasonal effect:** peak season raises volume and changes the operating regime.

S5(d). Which operational indicators should be monitored after deployment?

Recognition confidence and failure rate by package type (especially reflective); abstention/unknown-object rate; route length, travel time and delivery time vs planned; frequency of re-planning and blocked paths; safety-stop and near-miss counts; congestion and throughput (packages per hour); collision-risk events near workers; human-review volume for damaged packages and override rates; map-version/layout-change logs; operational health (latency, battery, uptime); and input drift statistics of camera images (lighting, reflectance).

S5(e). Under what conditions should the robot abstain, stop, use a fallback procedure, or request human review?

- **Abstain:** when recognition confidence for a package is below a threshold (for example a reflective package it cannot classify), do not pick or sort it; flag it.
- **Stop:** when the safety controller detects high collision risk or a person in its path, or when entering a restricted area is possible; safety rules always override the learned policy.
- **Fallback:** when the map does not match observed layout (repeated failed routes, blocked paths), switch to a conservative mode (lower speed, pre-approved safe routes, re-mapping) or a manual dispatch procedure; roll back to a previous model if a new version increases failures.
- **Request human review:** damaged packages, unrecognized or unusual objects, repeated recognition failures, and ambiguous situations near workers.

Part 7: Reflection Questions R1 to R5

The notes say these have "no single correct answers" and cannot be answered in a couple of sentences. The answers below give a complete, defensible structure. In a quiz, use the headings as your skeleton and add two or three points under each.

R1: National Early Warning System for Disease Outbreaks

R1(a). Reformulate the problem comprehensively and choose the problem type.

Worksheet statement: "Every Monday (index time), for each district in the country (unit of analysis: district-week), using only data received and validated by Sunday midnight (hospital admissions for respiratory and gastrointestinal syndromes, laboratory test orders and positivity, pharmacy sales, school absenteeism, weather, search and social-media signals, each with its reporting lag), produce (i) a calibrated probability that confirmed cases of a target disease will exceed the district's seasonal epidemic threshold within the next 2 to 4 weeks, (ii) a forecast with prediction intervals of cases and hospital admissions, and (iii) an anomaly flag for unusual syndromic patterns, so that the national public-health agency's epidemiology team can decide whether to issue an alert, dispatch investigation teams, pre-position test kits, or notify hospitals to prepare surge capacity. The system will be compared with the existing surveillance system and seasonal baselines and evaluated with detection delay, false-alarm rate, sensitivity, calibration, and regional and subgroup performance, while accounting for the cost of missed and late outbreaks, alert fatigue and public panic, data inequality between urban and rural areas, investigation-team capacity, and privacy governance."

- **Target population:** the national population, represented by districts; include rural and urban districts; separate pathogens (for example influenza-like illness, cholera, dengue) because signals differ.
- **Unit of analysis:** district-week (not individuals), which keeps privacy and matches the response unit.
- **Index time:** weekly (daily during active alerts), after data arrive and pass validation.
- **Prediction horizon:** 2 to 4 weeks ahead, the lead time needed to mobilize staff, supplies and hospital beds (a horizon too short is useless, like the six-minute flood warning).
- **Output:** probabilities, forecast intervals and anomaly flags with uncertainty, plus abstention when data are too incomplete.
- **Decision-maker:** the national epidemiology/surveillance unit, with regional health officers; not an automatic public announcement.
- **Intended intervention:** field investigation, confirmatory testing, hospital preparedness, targeted communication, vaccination or supplies.
- **Resource constraints:** limited investigation teams and test kits; limited ability to respond to many alerts at once.
- **Acceptable delay:** alerts must arrive before hospitals are overwhelmed (for example at least two weeks earlier than the current surveillance signal).

Problem type: hybrid. Forecasting (future admissions with intervals, temporal ordering), anomaly detection (new or unexpected patterns, where "unusual is not the same as undesirable" and an investigation policy is required), classification (probability of exceeding a threshold), and decision optimization (allocating limited teams). Causal estimation is needed later to judge whether interventions worked. A single classifier would ignore time, uncertainty and capacity.

R1(b). Examine the meaning of the available data (proxies, missing populations, selective labels, measurement bias).

Source	Directly measures	Only approximates	Bias and variation
Hospital admissions	People admitted with certain diagnoses	Community disease activity (severe cases only)	Access to hospitals differs; coding practices vary; lag; late-stage signal
Laboratory test orders/results	Tests ordered and positives	Infection prevalence	Testing much more frequent in wealthy urban areas; rural areas look "healthy" because unobserved (selective labels, access bias)
Pharmacy sales	Purchases of medicines	Illness in the community	Includes preventive purchases, stockpiling after news, promotions; pharmacy coverage uneven
School absenteeism	Children absent from school	Illness among children	Holidays, weather, strikes; covers only school-age children
Internet search	Search queries	Symptoms and concern	Changes after media reports; platform changes (Google Flu Trends); requires internet access
Social media	Posts by users	Public concern and symptoms	Concentrated among younger, urban users; bots; rumours
Weather	Temperature, rainfall, humidity	Conditions favouring transmission	Station coverage; a driver, not a measure of disease

- **Proxies:** almost every source is a proxy for disease; their relationship to true prevalence can change (proxy instability).
- **Missing populations:** rural, poor, elderly, offline populations and people who do not seek care.
- **Selective labels:** confirmed cases exist only where tests were done; negatives in low-testing areas mean "not observed", not "no disease" (the meaning of "negative").
- **Measurement bias:** case definitions, lab capacity and reporting lags differ by region and change over time.

R1(c). Design an evaluation strategy that matches the deployment claims.

- **Temporal validation:** train on earlier seasons and test on later ones (rolling-origin validation), because the claim is about predicting future outbreaks. A random train-test split would leak future information (weeks from the same outbreak on both sides, autocorrelated data) and make performance look better than reality.
- **Geographic validation:** hold out entire regions (especially rural, low-testing regions) to test transportability.
- **Calibration:** check that a predicted 30% chance of exceeding the threshold matches the observed frequency; check interval coverage for forecasts.
- **Subgroup/regional analysis:** sensitivity, false-alarm rate and delay by region, urban vs rural, income level and age group.
- **Detection delay:** how many weeks earlier than the current surveillance system the alert fires (the key utility metric).
- **False-alarm rate:** alerts per region per season, because false alarms cause alert fatigue, wasted teams and public panic.
- **Baseline comparison:** compare with existing epidemiological surveillance and seasonal/naive baselines; Google Flu Trends showed that combining new signals with lagged official data beat either alone.
- **Stress tests:** high-media periods, new pathogens, data outages.
- **Prospective silent evaluation** before any public use.

R1(d). Analyze the feedback loop created by deployment and propose a logging, monitoring and retraining strategy.

An alert changes the data-generating process: more testing is ordered (more positives found), more people go to hospital, searches and social-media posts increase, pharmacies sell more. The next week's data therefore look worse partly because of the alert (a performative effect), and retraining on these data would teach the model that its alert was correct (self-reinforcing bias). Conversely, a successful intervention reduces cases and may make the alert look like a false alarm.

Strategy:

- **Log every alert and response** (counterfactual logging): which districts were alerted, when, what actions were taken, testing volumes before and after.
- **Adjust for testing effort:** track test positivity rate and tests per capita, not just case counts; use lab-confirmed cases and hospital admissions as more stable anchors.
- **Mark post-alert periods** in training data, or model them explicitly, so the model does not learn from its own influence.
- **Keep held-out comparison areas** or use staggered rollout where ethically acceptable to estimate the alert's effect.
- **Monitor** input drift, residual error, calibration, alert volume, media events and platform changes; persistent residual growth should trigger recalibration or suspension (Google Flu Trends lesson).
- **Retrain on a schedule with human epidemiologist review**, not automatically after each alert.

R1(e). Develop a governance plan using Govern, Map, Measure and Manage.

Responsibilities: public-health officials (own the decision to warn and the response); epidemiologists (define case definitions and thresholds, review every alert); data engineers (pipelines, validation, lineage, privacy); model developers (models, calibration, documentation, model card); hospitals (timely, consistent reporting, surge planning); communication teams (clear public messages that avoid panic); government leaders (mandate, resources, legal basis, accountability).

- **Govern:** a surveillance governance board chaired by the national public-health authority; a named accountable owner; privacy rules (aggregate district data, no individual social-media tracking), data-sharing agreements, documentation (datasheets, model card), and escalation rules.
- **Map:** identify risks: under-detection in rural/low-testing areas, false alarms and panic, stigma of alerted districts, privacy harms, proxy instability, feedback loops.
- **Measure:** detection delay, false-alarm rate, calibration, regional and subgroup performance, data completeness, feedback indicators.
- **Manage:** decision rules for action:
  - **Issue a warning** when the calibrated probability exceeds the agreed threshold and an epidemiologist confirms with at least one anchored source (lab or hospital data).
  - **Abstain** when key sources are missing or delayed (for example a district's lab reports absent), and show "insufficient data".
  - **Request expert review** for anomalies driven mainly by search or social-media signals, or for new patterns.
  - **Suspend** the model if residual error or false alarms exceed limits, if a data-source definition changes, or if the system is shown to systematically miss certain regions; fall back to traditional surveillance.

R2: Adaptive Traffic Management for a Large City

R2(a). Critique the objective function and propose a multi-objective formulation.

Minimizing average vehicle travel time is mathematically clear but socially incomplete: it counts only vehicles (not people), weights every vehicle equally (a bus with 60 passengers counts like a car with one), ignores pedestrians and cyclists, ignores safety, pollution and where traffic goes, and ignores the distribution of costs. Optimization is "ethically blind"; it found the most efficient path to the specified objective by pushing costs onto buses, pedestrians, cyclists and low-income neighbourhoods. The 12% improvement is a success on a proxy objective.

Multi-objective formulation (minimize over signal plans):

$$J = w1\cdot(\text{person-weighted travel time}) + w2\cdot(\text{bus delay and schedule deviation}) + w3\cdot(\text{pedestrian waiting time}) + w4\cdot(\text{cyclist risk}) + w5\cdot(\text{emergency response delay}) + w6\cdot(\text{emissions}) + w7\cdot(\text{neighbourhood through-traffic})$$

Subject to hard constraints: maximum pedestrian wait (for example ≤ 60 s), minimum safe crossing times, emergency-vehicle priority, speed and safety rules, and **equity constraints** (no neighbourhood's through-traffic or pollution may increase by more than an agreed limit). Weights should be set by the city through public consultation, not by engineers alone, and results reported by group and area.

R2(b). Distinguish prediction, planning, optimization and control within this system.

- **Prediction/forecasting:** forecasting traffic volumes, congestion and bus arrivals over the next minutes to hours from cameras, sensors, phone data and history.
- **Planning:** choosing sequences of signal plans and coordination across corridors (green waves), and planning routes for emergency vehicles.
- **Optimization:** selecting signal timings that minimize the chosen objective under constraints.
- **Control:** continuously observing the network and adjusting signals in real time: x_{t+1} = F(x_t, u_t, w_t); the actions change the traffic being predicted.

It cannot be described as a traffic-prediction model because its signals **act** on the city: it changes routes, travel behaviour and neighbourhood traffic, creating feedback loops. Harm came from the optimization objective and control actions, not from prediction error.

R2(c). Analyze the representation of the city: who is visible and who is poorly represented?

- **Vehicles are highly visible** (cameras and sensors at intersections); **pedestrians and cyclists** are poorly represented (sensors often detect only vehicles; camera placement on major roads).
- **Mobile-phone location data** over-represent people who own smartphones and share location (younger, wealthier); under-represent elderly and poor residents; phone density can be mistaken for demand (proxy encoding).
- **Camera and sensor coverage** is uneven, often concentrated on arterial roads and wealthy districts; low-income areas may have fewer sensors, so extra traffic there is not seen.
- **Historical planning decisions** (which roads were built wide, where sensors were installed) are embedded in the data; optimizing on them reproduces past priorities.
- **Aggregation bias:** citywide average travel time hides that some groups (bus riders, pedestrians, residents of certain neighbourhoods) are worse off.

R2(d). Propose a decision-cost framework.

$$C(t) = c_FP \cdot P(\text{false congestion alarm}) + c_FN \cdot P(\text{missed congestion}) + c_emerg \cdot E[\text{emergency delay}] + c_ped \cdot (\text{pedestrian risk}) + c_poll \cdot (\text{pollution}) + c_workflow$$

- **False congestion alarms:** unnecessary signal changes, diverted traffic into neighbourhoods, wasted green time.
- **Missed congestion:** gridlock, delays, spillback.
- **Delayed emergency vehicles:** very high cost (lives), so emergency priority is a hard constraint.
- **Pedestrian risk:** injury and death; hard safety constraints on crossing times.
- **Pollution:** health costs concentrated in exposed neighbourhoods.
- **Workflow failure:** operators overwhelmed, manual overrides, system outages.

These costs should decide real-time control actions (for example do not shorten pedestrian phases even if it improves average travel time), rather than average accuracy or travel time alone.

R2(e). Treat the system as a socio-technical composite and decide who may modify objectives, thresholds and safety constraints.

- **Model:** forecasting and optimization/control algorithms.
- **Data infrastructure:** cameras, sensors, phone data, transit feeds, their coverage and quality.
- **Public interface:** signals, public dashboards, travel apps, and channels for complaints.
- **Traffic-control workflow:** operators who monitor, override and respond to incidents.
- **Transport policy:** priorities for public transport, walking, cycling, safety (Vision Zero).
- **Legal rules:** traffic law, safety standards, environmental and equity law, data protection.
- **Accountability:** council oversight, audits, public reporting.

Authority: objectives and weights should be set by the elected city council/transport authority after public consultation; safety constraints by the transport safety authority and traffic engineers (and they cannot be traded away by the optimizer); thresholds and operational parameters by the traffic-control centre within approved limits, with changes versioned and logged; independent auditors review equity impacts. Engineers implement but should not alone define what "success" means.

R3: Generative AI Adviser for University Students

R3(a). Locate the system in the AI landscape and develop a hybrid architecture.

Component	Role and appropriate use
Generative AI (LLM)	Natural-language conversation, explanation and drafting of study plans; should not be the source of facts.
Information retrieval	Retrieve current curriculum documents, course offerings, policies (retrieval-augmented generation) so answers are grounded.
Symbolic rules	Prerequisite checking, degree-requirement audits, credit limits, eligibility rules; deterministic and verifiable.
Search	Search over the space of course combinations and schedules that satisfy constraints.
Recommendation	Rank suitable courses for the student's goals and history (with care about feedback loops from historical enrolment).
Optimization	Build timetables that satisfy constraints (clashes, workload).

Component	Role and appropriate use
Automation	Routine actions such as booking an appointment with a faculty adviser or saving a draft plan.
Human decision support	Faculty advisers approve study plans and handle exceptions.

Hybrid design: the LLM is only the interface. Facts come from retrieval over authoritative, versioned sources; every recommendation passes a rule-based prerequisite and requirement checker; schedule options come from search/optimization; outputs are shown as a draft with citations; registration and exceptions go to a human adviser. Student records are accessed only through an access-controlled service that returns the current user's data.

R3(b). Define the boundary between intelligence and autonomy.

Fluency is not authority: "autonomy must not be conflated with intelligence." Levels by risk:

- **Passive/informational (allowed):** explaining policies, describing courses with citations.
- **Decision support (default):** proposing course lists and study plans that the student and adviser review.
- **Semi-autonomous (low-risk actions only):** booking an advising appointment, saving a plan, sending reminders, with undo.
- **Not permitted without human approval:** registering or dropping courses, changing degree programmes, anything affecting graduation, fees, visas, scholarships or academic standing.

Automatic registration is inappropriate now, because the pilot shows invented prerequisites, unavailable classes, outdated rules and privacy leaks; errors would be costly and hard to reverse (missed prerequisites delay graduation). If ever introduced, it should be semi-autonomous: only rule-verified, low-risk registrations, with student confirmation, an adviser sign-off queue and easy reversal.

R3(c). Design mechanisms for grounding and reliability.

- **Retrieval from authoritative sources:** answers generated only from the current, official catalogue, timetable and regulations.
- **Rule-based prerequisite checking:** every recommended course is checked by a deterministic rule engine against the student's record and current offerings.
- **Source citation:** every factual statement links to the document and version used.
- **Uncertainty communication:** state confidence and when information may be incomplete.
- **Abstention:** "I cannot confirm this; please contact your adviser" when no authoritative source is found or sources conflict.
- **Version control:** documents and rules versioned with effective dates; outdated regulations retired; the model and prompts versioned.
- **Human escalation:** exceptions, conflicting rules, high-stakes questions go to a faculty adviser.

Together these reduce confabulation (NIST's first generative risk) and outdated advice.

R3(d). Examine human-system interaction risks and propose interface features.

Risks: **automation bias** and **overreliance** (students follow fluent advice without checking); **interface design** that presents answers as authoritative; **information integrity** (professionally written but incorrect advice accepted without verification, like the junior associate and the NDA); **unequal access** (students with fewer human advisers or less confidence rely more on the AI, so errors fall unevenly); privacy fears.

Interface features: visible citations and "last updated" dates; a clear banner that advice is not official until approved; confidence indicators; highlighting any requirement the checker could not verify; a "check with

adviser" button and easy booking; showing alternatives rather than one answer; requiring confirmation of key facts before any action; reporting of wrong answers; and guaranteed human-adviser availability for disadvantaged students.

R3(e). Develop an end-to-end monitoring plan and rollback triggers.

Indicator	How measured
Factual accuracy	Audited sample of answers vs official sources; hallucinated prerequisites or courses
Privacy violations	Any exposure of another student's data (target: zero); access-log audits
Outdated knowledge	Answers citing superseded document versions
Student overrides	How often students or advisers reject recommendations
Human escalations	Volume and reasons for referral to advisers
Harmful recommendations	Plans that would delay graduation or break rules
Subgroup differences	Accuracy, escalation and outcome by programme, year, international status, first-generation students
Latency and availability	Response time, uptime
Real-world outcomes	Registration errors, graduation delays, satisfaction, adviser workload

Triggers: any confirmed exposure of another student's record → immediate suspension of record access and incident response; a rise in hallucinated prerequisites or outdated-regulation answers above an agreed rate → rollback to the previous version or retrieval-only mode; policy changes not yet ingested → temporary suspension of affected topics; subgroup gaps or harmful plans above limits → restrict to informational mode until fixed.

R4: AI for Climate-Smart Agricultural Support

R4(a). Separate prediction from causal reasoning.

Predicting yield accurately answers "What is likely to happen?" Recommending irrigation or fertilizer requires "What would happen if the farmer did X instead of Y?" (a causal question). The yield model learned associations from commercial farms, where irrigation and fertilizer use are confounded with wealth, soil quality, seed varieties and management skill; high-yield farms irrigate more, but that does not prove irrigation causes the extra yield on a smallholder plot (like attendance and exam scores). Counterfactual questions to answer:

- Would this farmer's yield be higher if they irrigated on day d rather than day d + 5, given their water access?
- Would applying fertilizer amount a at time t increase yield on this soil and crop variety, and by how much relative to its cost?
- Would changing the planting date by two weeks increase yield under this season's uncertain rainfall?
- What is the effect for smallholders specifically, not averaged over commercial farms?

These need causal estimation $\tau = E[Y(1) - Y(0)]$ with field trials (randomized plots), agronomic mechanistic crop models, and careful control of confounding.

R4(b). Evaluate the target population and data representativeness.

Training population: large commercial farms with reliable irrigation, good sensors, stable internet, standard varieties. Deployment population: smallholders with incomplete records, different varieties, irregular water

and limited capacity. This is **distribution shift** (population, device and context). **Missingness** is not random (small farms lack sensors and records, so missing data are correlated with being poor). **Sensor inequality** means features available in training are absent or noisier in deployment (train-serving mismatch). **Regional variation** in soils, climate and pests is likely under-represented. A **Datasheet for Datasets** would reveal all this before deployment by documenting composition (farm types, regions, varieties), collection process (sensor types), missing populations (smallholders) and recommended uses, just as the skin-lesion datasheet would have revealed the skin-tone gap.

R4(c). Design a meaningful representation of the farming process.

Static feature vectors (one row per farm-season) are insufficient: they cause **loss of context**. Farming is temporal and spatial: timing of rainfall relative to planting, soil moisture over time, growth stages. A good representation includes **temporal sequences** (daily weather, soil moisture, crop stage), **spatial maps** (satellite imagery, soil maps, neighbouring plots), **weather uncertainty** (probabilistic forecasts, not single values), **resource constraints** (water availability, labour, credit, fertilizer access) and **mechanistic crop models** that encode known plant physiology. Losing temporal context can make the model recommend irrigation after the critical window; losing local context can apply commercial-farm patterns to very different plots.

R4(d). Choose an appropriate combination of computational methods.

- **Statistical learning:** satellite-based crop monitoring, pest and disease detection, yield estimation.
- **Weather forecasting:** probabilistic short- and seasonal forecasts.
- **Mechanistic crop models:** simulate how water and nutrients affect growth, filling gaps where smallholder data are scarce.
- **Rule-based agronomic constraints:** do not recommend irrigation when no water is available; fertilizer limits; local regulations.
- **Causal estimation:** field trials to estimate the benefit of each recommendation for smallholders.
- **Optimization:** choose the best feasible action given the farmer's budget, water and labour.
- **Human expert review:** extension officers validate recommendations and handle unusual cases.

A single complex black-box model trained on commercial farms may be inferior: it cannot encode constraints, extrapolates poorly under shift, cannot explain itself to farmers or extension workers, and confuses prediction with intervention effect. A transparent hybrid is more robust, auditable and trusted.

R4(e). Assess the broader decision workflow.

A recommendation is useful only if the farmer can act on it: water, credit, fertilizer, labour, transport and market access. Therefore:

- **Problem formulation:** the decision is "what is the best feasible action for this farmer given their resources", not "maximize yield"; the objective may be income or risk reduction, not yield.
- **Evaluation:** measure adoption, feasibility, farmer income, yield variance and harm (debt from inputs), by farm size and region; compare with extension-officer advice (baseline).
- **Interface design:** low-bandwidth, local language, voice or SMS, offline mode, explanations and uncertainty, and options at different costs.
- **Claims of success:** do not claim impact from accurate yield prediction; claim only what field evidence shows, and report where the system does not work.

R5: Automated Prioritization of Social-Welfare Applications

R5(a). Does the label represent the real objective?

No. The label is "approved historically", but the real objective is present need or eligibility. Historical approvals encode **policy choices** (past rules and quotas), **documentation inequality** (applicants with complete records were easier to approve), **resource constraints** (limited staff approved easy cases), and **institutional bias** (caseworker discretion, stereotypes). Informal-sector workers and remote applicants had incomplete records, so they were approved less, not because they were less needy. The model reproduces "historical, potentially biased institutional decisions rather than discovering an objective, optimal truth" (status quo bias). This mirrors the Obermeyer cost-as-proxy example.

R5(b). Analyze the emerging feedback loop.

Incomplete records → low-priority queue → reviewed less often and later → fewer approvals observed for these groups (many never get a decision, or give up) → these outcomes become new labels (**selective labels**: we only see outcomes for cases that were reviewed) → retraining learns an even stronger link between incomplete records, informal work, remote location and non-approval → the model becomes increasingly confident in its original decisions (self-reinforcing bias). The model is judged on outcomes it produced, so it looks more accurate while harming the most vulnerable.

R5(c). Reformulate the problem around a decision or capability.

- **Real operational need:** process applications so that people in urgent need receive assistance quickly and eligible applicants are not wrongly delayed, given limited staff.
- **Target population:** all applicants in the current period; unit: one application.
- **Index time:** at submission (and again when missing documents arrive).
- **Prediction horizon:** harm from delay over the next 2 to 4 weeks.
- **Decision alternatives:** fast-track, standard review, request documents with assistance, outreach visit.
- **Capacity constraints:** caseworker hours per week.
- **Consequences of delay:** hunger, eviction, loss of medical care; delay is itself harm.

Prioritization should be based on **urgency and need** (for example housing loss, dependants, no income), combined with **review complexity** (incomplete records get help, not lower priority), not on predicted historical approval. Eligibility uncertainty should trigger assistance and human review, not demotion. A transparent rule-based urgency score may be better than an opaque model.

R5(d). Design an evaluation and fairness framework.

- **False-negative rate** (truly needy applicants placed in low priority) overall and by group (informal sector, remote, gender, disability, ethnicity).
- **Waiting time** to decision, by group and queue.
- **Calibration** of any urgency/need probability by group.
- **Subgroup performance** and intersections (remote × informal).
- **Appeal outcomes:** share of appeals upheld, by group.
- **Human override rates:** how often caseworkers move cases between queues and why.
- **Harm from delayed assistance:** evictions, debt, health events.
- **Incomplete ground truth:** outcomes are missing for unreviewed cases. Handle by randomly auditing a sample of low-priority cases (reviewing them fully to learn true eligibility), counterfactual logging of queue assignment, and not treating unreviewed as ineligible (the meaning of "negative").

R5(e). Assign responsibility across the socio-technical lifecycle and propose safeguards.

Actor	Responsibility
Policy designers	Define the objective (need, urgency) and eligibility rules; decide what the system may do
Data collectors	Make record-keeping accessible; record why data are missing; avoid penalizing informal work
Software developers	Build interfaces and pipelines; log decisions; enforce access control and privacy
Model developers	Choose appropriate labels, test for proxies (location, employment type), calibrate, document a model card
Caseworkers	Review and override priorities; record reasons; flag errors
Agency leaders	Accountable owner; approve deployment; fund capacity; suspend if harmful
Auditors	Independent fairness and performance audits, including random audits of low-priority cases
Appeal bodies	Hear challenges and feed results back into monitoring

Safeguards: human review for every low-priority assignment; explanation of the priority and missing documents to applicants; an appeal channel with a response time; abstention for incomplete records (route to assisted review rather than low priority); monitoring by subgroup, waiting time and appeal outcomes; no automatic retraining on biased decisions (counterfactual logging, audits); and decommissioning criteria (if disparities are not reduced within an agreed period, or if the label cannot be fixed, stop using the model and revert to need-based rules).

Part 8: Application Problems A1 to A7

These are the assignment-style tasks. Your Assignment 1 already had four of these types (worksheet, model card, datasheet, failure analysis), so the quiz may ask you to produce a short version of any of them. Each answer below is complete; for the quiz, keep the headings and shorten the bullets.

A1: Formulating an AI Problem (Reducing Unplanned Hospital Readmissions)

> **A1. Develop a complete AI problem formulation for the readmission project and identify variables that could cause temporal leakage.**

Element	Specification
Real-world decision	Which discharged adult patients should receive one of the 100 weekly early follow-up appointments, a telephone call, standard discharge instructions, or no additional intervention, so that preventable unplanned readmissions fall.
Target population and exclusions	Adult (18+) patients discharged alive from medical and surgical wards. Exclude: planned readmissions (scheduled chemotherapy, staged surgery), patients discharged to hospice or palliative care, transfers to another acute hospital, patients who leave against medical advice (handled separately), obstetric admissions, and deaths in hospital.
Unit of analysis	One index hospitalization (discharge episode). Patients with several admissions are kept in one data partition (patient-level split) to avoid group leakage.
Index time	At discharge: when the discharge order is signed, before the patient leaves (so the discharge team can book the follow-up).
Prediction horizon	30 days after discharge.
Outcome definition	Y = 1 if the patient has an unplanned (emergency or urgent) inpatient readmission to any hospital in the network within 30 days of discharge; planned readmissions and ED visits without admission are not counted. Deaths within 30 days are recorded as a separate outcome, not coded as "no readmission".
Information available at index time	Demographics; diagnoses and procedures of the current stay; medications at discharge; laboratory results finalized before discharge; number of previous admissions and ED visits in the past 12 months; length of stay; appointment history before discharge (for example previous no-shows); insurance status as recorded at admission; discharge destination.
Decision-maker	The discharge team (discharge nurse/coordinator with the attending physician).
Actions	Tier 1 (highest risk, within capacity): early follow-up appointment within 7 days. Tier 2: structured telephone call within 48 to 72 hours. Tier 3: standard discharge instructions. Low risk: no additional intervention.
Baseline / status quo	Current clinician judgment and existing practice (for example follow-up based on discharge nurse assessment), plus a simple heuristic score (such as the LACE index: length of stay, acuity, comorbidity, ED visits) as a second comparator.
False positives	A low-risk patient takes one of the 100 slots, displacing someone who needed it; wasted clinic time and patient inconvenience.
False negatives	A high-risk patient receives only standard instructions, deteriorates at home and is readmitted (or dies); more serious and less reversible, so c_FN > c_FP.
Intervention-capacity constraint	Exactly 100 follow-up appointment slots per week; telephone-call capacity set by nursing staff hours. The decision rule is ranking: the top 100 by $\hat{p}$ each week (k = 100) after review; the next band receives calls.

Element	Specification
Primary performance and calibration measures	Primary: precision@100 and recall@100 (share of readmissions captured by the slots). Secondary: AUROC (discrimination), calibration curve and calibration slope, positive predictive value, number of preventable readmissions averted in a pilot.
Subgroup and fairness analyses	Age groups, sex, ethnicity, primary language, insurance type, socioeconomic area, diagnosis group, and discharge destination; report selection rates and recall per group with confidence intervals. Check whether insurance status acts as a proxy that lowers priority for uninsured patients (access bias).
Abstention / human review	Abstain and send to clinician review when key inputs are missing (for example no discharge medications recorded), when the patient is in a group poorly represented in training, or when $\hat{p}$ lies in an uncertainty band around the slot cut-off. Clinicians can override with a logged reason.

Final problem statement (one paragraph): At the time the discharge order is signed, for every adult medical or surgical patient discharged alive (excluding planned readmissions, hospice discharges and transfers), using only information recorded and available before discharge, estimate the calibrated probability of an unplanned inpatient readmission within 30 days, and rank patients each week so that the discharge team can allocate the 100 early follow-up appointments to the highest-ranked patients after clinical review and offer telephone follow-up to the next risk band. The system will be compared with current clinician judgment and the LACE index and evaluated using precision@100, recall@100, calibration and discrimination, reported overall and by age, sex, ethnicity, language, insurance and diagnosis group, while accounting for the higher cost of missed readmissions, the fixed appointment capacity, abstention for incomplete records, and governance requirements including a model card and post-deployment monitoring.

Temporal leakage variables:

- **Future claims data:** claims after discharge reveal the readmission itself (target leakage).
- **Appointment history after discharge** (whether they attended follow-up), which also reflects the intervention.
- **Laboratory results finalized after discharge** (pending labs).
- **Discharge summaries completed or edited after discharge** (often written days later and may mention complications).
- **Diagnoses coded after discharge** for billing (ICD codes finalized later).
- **Predictions generated "several days after discharge"** would use post-discharge information and leave little time to act; that option is rejected.
- **Insurance status updated later** (only the value at admission/discharge can be used).

A2: Preparing a Model Card (Heart Disease Risk Prediction Model)

This Model Card follows the structure of the notes' "Instructive Example of a Model Card" (Section 4.8.1): Model Overview, Data Description, Performance Metrics, Known Limitations, Intended and Inappropriate Uses, Ethical Considerations, Additional Considerations, then Takeaways. Extra items required by the task (calibration, human review, monitoring, triggers, recommendation) are added inside the most relevant section.

> **A2. Prepare a complete Model Card for the heart disease system and give a deployment recommendation.**

Model Card: Heart Disease Risk Prediction Model

Model Overview

- **Model Name:** Heart Disease Risk Prediction Model.
- **Version:** 1.0 (first evaluated release; internal validation only).
- **Type:** Binary classification (clinically significant heart disease: Yes/No), producing a probability of heart disease.
- **Model Architecture:** Random Forest Classifier trained on structured patient data (age, sex, blood pressure, cholesterol, chest pain category, maximum heart rate, diabetes status, BMI, ECG findings).
- **Purpose:** To estimate the probability that an adult patient has clinically significant heart disease based on clinical measurements.
- **Intended clinical role:** Clinical decision support. The model is intended to assist healthcare professionals by providing an additional tool to help identify higher-risk patients who may require further diagnostic testing (for example stress testing or angiography) or preventive care. It does not make or execute decisions.
- **Intended users:** Physicians, cardiologists and trained clinical staff in the two developing hospitals (and similar urban hospitals after local validation). Not patients, insurers or administrators acting without a clinician.

Data Description

- **Training data:** 12,000 adult patient records from two urban hospitals.
- **Evaluation data:** 3,000 records collected later from the same two hospitals (a temporal holdout from the same sites; this is not external validation).
- **Features (inputs):** age, sex, blood pressure, cholesterol, chest pain category, maximum heart rate, diabetes status, body mass index, and electrocardiogram findings.
- **Output:** a probability (0 to 1) of clinically significant heart disease.
- **Data Preprocessing and missing-value handling:** not stated in the scenario and must be documented. Recommended (following the notes' example): missing numerical values imputed with the mean or median and categorical values with the mode, using the same versioned transformation in training and serving (to avoid train-serving mismatch); a missing-value indicator retained; records with missing critical inputs (for example ECG findings) flagged for abstention rather than silently imputed.
- **Exclusions:** the model does not include family history, smoking status, ethnicity or genetic information, as these were not available in the dataset.

Performance Metrics

- **Overall Accuracy:** 86%
- **Precision:** 0.82 (82% of patients flagged as positive truly have heart disease)
- **Recall:** 0.74 (the model misses about 26% of patients who have heart disease)
- **F1 Score:** 0.78
- **Performance by Subgroup:**
  - Male patients: Recall = 0.80
  - Female patients: Recall = 0.62 (about 38% of women with heart disease are missed)
  - Patients younger than 40 years: Recall = 0.55 (about 45% missed)
  - Not reported and required: precision by subgroup, older adults, diabetic vs non-diabetic patients, ethnicity (unavailable), and intersections (for example young women).
- **Calibration findings:** the predicted probabilities are moderately overconfident: when the model outputs a high probability, the observed rate of heart disease is lower (and low probabilities may be too low). Probabilities should not be read literally until the model is recalibrated (for example Platt scaling or isotonic regression on held-out data) and a calibration curve is published overall and by subgroup.

- **Fairness Observations:** The model performs substantially better on male patients than female patients (recall 0.80 vs 0.62), indicating potential gender bias, and performs worst for patients under 40 (recall 0.55). The model therefore tends to miss heart disease in women and younger adults, groups whose symptoms are often atypical and who are already at risk of under-diagnosis.

Known Limitations

- **Bias in Data:** the dataset likely contains fewer female and younger patients with heart disease, and women often present with atypical chest pain categories, leading to weaker performance for these groups.
- **Generalizability:** the model was trained and evaluated only on two urban hospitals and has not been externally validated. Performance may differ in rural hospitals, other regions, other countries, primary care or populations with different demographics and equipment.
- **Feature Limitations:** family history, smoking status, ethnicity and genetic information were unavailable; these are important risk factors, which may reduce accuracy and hide ethnic disparities that cannot be measured.
- **Calibration:** probabilities are moderately overconfident.
- **Recall:** overall recall of 0.74 means one in four patients with disease is missed; this is too low for the model to be used to rule patients out.

Intended and Inappropriate Uses

- **Intended (appropriate) Use:** to support clinicians in identifying patients at higher risk of heart disease who may benefit from further diagnostic testing or preventive care. It should be used as an additional decision support tool alongside clinical judgment and standard risk tools, never as the sole basis for diagnosis.
- **Inappropriate or prohibited Uses:**
    - **Automatically rejecting requests for further cardiac testing** (as some administrators proposed). This is prohibited: with recall of 0.74 overall, 0.62 for women and 0.55 for under-40s, a "low-risk" output would deny testing to many patients who actually have heart disease.
    - Using the model as the sole basis for diagnosis or for ruling out disease.
    - Use for populations not represented in training (other regions, rural settings, children) without local validation.
    - Use by insurers or employers for pricing, eligibility or employment decisions.
    - Using the probability as a precise value before recalibration.

Ethical Considerations

- **Fairness:** the model has potential gender and age biases, which should be addressed before full-scale deployment. Regular audits should be conducted to ensure fair treatment of all patient subgroups; thresholds should not produce much higher false-negative rates for women or younger patients.
- **Harms from false negatives:** a patient with heart disease is classified as low risk, so testing and treatment may be delayed, risking myocardial infarction or death. Because false negatives are concentrated in women and younger adults, the harm is unequal and could exacerbate health inequalities (the notes: "under-predicting heart disease in women could delay necessary interventions").
- **Harms from false positives:** unnecessary invasive tests (angiography carries bleeding and infection risk), anxiety, cost, and wasted diagnostic capacity.
- **Human-review requirements:** a clinician must review every output; the model can prompt further testing but must never block it. Low-probability outputs for patients with symptoms, women, under-40s, or incomplete records must receive full clinical assessment. Clinicians should be able to override the model with a logged reason, and outputs should be shown with uncertainty and the main contributing features.
- **Impact of Misuse:** using the model to automatically deny testing, or in under-represented populations, could cause serious harm and erode trust.

Additional Considerations

- **Generalizability limitations:** valid only for adult patients similar to those in the two urban hospitals; any new site must perform local external validation and calibration before use.
- **Monitoring requirements after deployment:** input quality and missingness; input drift (age, sex mix, new ECG machines, coding changes); output distribution and share of patients flagged; calibration on new outcomes; recall and precision overall and by sex, age group and site; clinician override and acceptance rates (automation bias); time to diagnosis for flagged vs non-flagged patients; adverse events among patients classified low risk.
- **Conditions that should trigger action:**
  - **Recalibration:** calibration error above the agreed limit overall or in any subgroup, or a change in disease prevalence.
  - **Retraining:** input drift (new population, equipment or coding), or subgroup recall gaps that persist after recalibration.
  - **Rollback:** a new version lowers recall for any subgroup or widens the male-female gap; rollback requires versioned models and data.
  - **Withdrawal:** evidence the model is being used to deny testing, adverse events linked to false negatives, or failure to reach an agreed minimum recall for women and younger patients within a set period.
- **Future Work:** integrate family history, smoking status and relevant genetic markers; collect a more diverse dataset (more women, younger patients, other ethnicities and rural hospitals); perform external validation at independent sites; recalibrate; examine threshold choice using the decision-cost function; and conduct a prospective silent evaluation before broader deployment.

Deployment Recommendation: Conditionally Approve (as clinical decision support only, within the two development hospitals).

Conditions: (1) recalibrate and publish calibration curves before use; (2) use a lower threshold or a separate review pathway so that recall for women and patients under 40 is raised to an agreed minimum; (3) the model may only recommend further testing and must never be used to automatically reject requests for cardiac testing; (4) every output is reviewed by a clinician, with overrides logged; (5) subgroup monitoring and the triggers above are in place at launch; (6) external validation before any use at other sites. If recall for women and younger patients cannot be improved, the recommendation becomes **do not approve** for general use.

Takeaways from the Model Card (as in the notes): **Data transparency** (users see where data came from and what is excluded); **Performance by group** (subgroup recall reveals the gender and age gaps that 86% accuracy hides); **Known limitations** (no external validation, missing risk factors, overconfidence); **Ethical guidance** (it supports but never replaces clinical judgment and must not be used to deny testing).

A3: Preparing a Datasheet for a Dataset (Crop Disease Image Dataset)

This follows the notes' Datasheet structure (motivation, composition, collection process, preprocessing, recommended uses, limitations) and your Assignment Problem 3.

> **A3. Prepare a Datasheet for the crop disease image dataset and decide whether it suits smallholder farmers using low-cost mobile phones.**

Motivation. Created to support research on AI systems that identify crop diseases from leaf images, enabling early, low-cost detection without waiting for a plant pathologist. The stated intended use is only "crop disease AI research", which is too vague and must be made specific.

Creators, funders and maintainers. The agricultural research institute's field and data teams, with 35 commercial partner farms. The funder must be stated. A named maintainer and contact address are required

(currently missing).

Composition.

- 80,000 images; 4 crops (wheat, maize, tomato, potato); 12 disease classes + 1 healthy class.
- **Unit of observation:** one leaf image with one crop and one class label. Several images may come from the same plant, plot or farm, so images are not independent.
- **Crop and disease distributions:** counts per crop and per class must be reported; diseased plants were photographed more often than healthy plants.

Geographic and seasonal coverage. 35 commercial farms; 70% of images from two regions; no images during drought; most images captured in daylight.

Data-collection process. Field technicians photographed leaves on site, mostly with high-resolution cameras; only 5% (about 4,000 images) with low-cost mobile phones.

Sampling procedure. Not random: farms chosen through partnership (all commercial, concentrated in two regions); within farms, symptom-driven (diseased plants over-sampled). Two selection effects: site selection and outcome-dependent selection.

Label definitions. One of 13 classes per image. Must document the definition of each disease, how co-infections and abiotic stress (nutrient, water, drought stress) were handled, and that severity/stage is not recorded.

Labelling and verification. Labels assigned by field technicians; only 10% independently reviewed by plant pathologists; the agreement rate must be reported. 90% rely on non-specialist judgment, so noise is likely in visually similar diseases.

Preprocessing and transformations. Resized to 512 × 512 (may distort lesions and remove absolute scale); backgrounds not removed; farm identifiers replaced with coded values; no colour normalization reported. Transformation code and versions must be published.

Known missing populations and conditions. Smallholder farms (entirely absent); drought conditions; regions outside the two main regions; low-light, overcast, rainy or dusty conditions; low-cost phone images; other crops and diseases; no "unknown/other" class.

Potential proxies and dataset artifacts. Background (soil, mulch, irrigation equipment, row spacing) linked to farm and disease; device characteristics; photographer framing (lesions framed centrally); lighting/time of day; resizing artifacts. A model may learn these shortcuts rather than leaf symptoms.

Class imbalance and selection bias. Healthy leaves are under-represented, so a model will over-diagnose disease in the field; rare diseases have fewer images and fewer verified labels; commercial and two-region dominance is selection bias.

Consent, privacy, licensing and access. Written agreements with each farm for photography, release and model training; workers visible in images must consent or be blurred; farm codes are a start but backgrounds and EXIF/GPS metadata can re-identify farms, so strip metadata and restrict access to the code key; an explicit licence stating commercial use, redistribution and attribution.

Recommended uses. Research and benchmarking with farm-grouped splits; pre-training/transfer learning before fine-tuning on local expert-labelled data; research on distribution shift, label noise and shortcut learning; prototype decision support for large commercial farms in the two well-represented regions with agronomist confirmation.

Uses that are not supported. Sole basis for diagnosis or pesticide decisions; deployment on smallholder farms; drought regions or seasons; crops or diseases outside the 13 classes; severity grading or yield-loss estimation; insurance or regulatory decisions; identifying individual farms.

Risks associated with mobile-phone deployment. Train-serving mismatch and device shift: phone images have lower resolution, stronger compression, different colour balance, blur and variable framing and lighting; fine disease distinctions depend on exactly the texture and colour detail phones lose. Accuracy may collapse while the model stays confident. False "diseased" → unnecessary pesticide costs; false "healthy" → outbreaks spread. Need a phone-specific validation set, calibrated confidence with abstention, image-quality checks, and a route to a human agronomist.

Generalization limitations. Performance should be expected to drop with new regions, smallholder settings, drought or other stress, different lighting, cultivars, seasons or devices, especially several at once.

Maintenance, correction, versioning and retirement. A named maintainer; a process to report and correct mislabelled images with a public changelog; fixed, reproducible versions (v1.0 remains available); a formal procedure to add new disease categories; planned additions targeting the gaps (smallholders, drought, phones, new regions); a retirement policy (for example when cultivars or pathogen strains change) with user notification.

Conclusion: not suitable on its own for a system for smallholder farmers using low-cost mobile phones. The target users (smallholders), the target device (phones: only 5% of data), and the target conditions (irregular water, possible drought, varied regions) are exactly what the dataset lacks; labels are mostly unverified and healthy plants are under-sampled. The model would be "confidently wrong rather than obviously broken". It can be used for pre-training, but a purpose-built dataset from smallholder fields, captured on farmers' phones, labelled by pathologists, across seasons, is required before deployment.

A4: Deciding Which Model to Deploy (Predictive Maintenance)

Table 3 in the notes (page 167) compares the three models:

Criterion	Model A: Logistic Regression	Model B: Random Forest	Model C: Deep Neural Network
Accuracy	88%	92%	94%
Failure recall	76%	87%	90%
Calibration	Good	Moderate	Poor
Inference latency	5 ms	35 ms	240 ms
Explanation	Direct coefficients	Feature importance and local explanations	Unstable post-hoc explanations
Memory requirement	Very low	Moderate	High
Specialist maintenance	Low	Moderate	High
Performance with missing sensors	Moderate	Good	Poor
External-site validation	Completed	Completed	Not completed
Retraining frequency	Every six months	Every three months	Every month

> **A4. Select the model that should be deployed and justify the decision (do not choose on accuracy alone).**

Decision: deploy Model B (Random Forest) as the primary model, in a hybrid arrangement, operating as decision support with an automatic safety layer. Reject Model C. Keep Model A as the interpretable fallback and benchmark.

Step 1: eliminate models that violate hard constraints. Model C's 240 ms latency breaks the controller's 100 ms requirement; the plant has no GPU and C needs high memory; its explanations are unstable, while engineers "require understandable reasons before stopping a machine"; it has poor calibration, poor performance with missing sensors, and no external validation; it needs monthly retraining and high specialist

maintenance, but the team has limited deep-learning expertise. Its extra 3 points of recall cannot compensate. Model C is out.

Step 2: compare A and B using error costs, not accuracy.

- **Failure recall:** B catches 87% of failures vs 76% for A. For every 100 real impending failures, B catches about 11 more.
- **False-positive vs false-negative costs:** a missed failure costs more than $1,000,000 and may injure workers; an unnecessary shutdown costs $20,000. So $c_{FN} / c_{FP} \geq 50$. Eleven extra caught failures are worth at least $11 million plus avoided injuries; B would have to cause more than 550 extra unnecessary shutdowns per 100 failures before it became worse than A. The notes' rule: the best threshold and model minimize real-world cost, $C(t) = c_{FP} \cdot P(FP|t) + c_{FN} \cdot P(FN|t) + c_{delay} \cdot E[D|t] + c_{workflow}(t)$, not maximize accuracy.
- **Predictive performance:** B is also more accurate (92% vs 88%).
- **Calibration:** A is better (good vs moderate). B's probabilities must be recalibrated (for example isotonic regression or Platt scaling on held-out data) before a threshold is set; calibration is checked after every retraining.
- **Latency:** both are well within 100 ms (5 ms and 35 ms), leaving room for data validation within the budget.
- **Hardware and memory:** B needs moderate memory and no GPU; runs on the existing CPUs.
- **Robustness to missing sensors:** B is good (A moderate); important because sensors fail in a factory.
- **Interpretability:** B gives feature importance and local explanations ("vibration in the last 30 min and bearing temperature raised the risk"), which is understandable enough for engineers; A is even clearer.
- **External validation:** completed for both.
- **Maintenance burden and team expertise:** B is moderate; the team has experience with statistical and tree-based models, so B is maintainable in-house.
- **Long-term lifecycle cost:** B requires retraining every three months (moderate), far less than C; A is cheapest but its lower recall costs far more in missed failures.
- **Monitoring and retraining:** monitor sensor data quality and missingness, input drift (new machines, loads, seasons), score distribution, alert rate and shutdowns, calibration, recall and precision as outcomes arrive (failures and maintenance findings), explanation stability, latency, and engineer override rates. Retrain every three months or when drift or recall drops trigger it; validate each new version against the current one and block release if recall or calibration worsens (TFX model validation); keep rollback.

Step 3: automatic operation or decision support? Decision support for normal alerts: stopping a machine costs $20,000, and engineers want reasons, so the model raises a prioritized alert with its probability and explanation, and an engineer decides (within a defined response time). However, for imminent-danger conditions, an explicit safety rule (symbolic, not learned) should stop the machine automatically, for example vibration or temperature above a hard engineering limit, because worker safety cannot wait for review. This is the semi-autonomous pattern: routine low-risk handling automated, ambiguous cases escalated.

Step 4: hybrid arrangement (preferable).

- **Model B** as the primary risk scorer (every minute), recalibrated, with a threshold set by $C(t)$ and by maintenance-crew capacity, plus alert suppression so the same machine does not alert every minute (avoid alert fatigue).
- **Model A** runs in parallel as an interpretable benchmark and **fallback**: if B's service fails, or if too many sensors are missing, the system falls back to A (or to rules) rather than giving no prediction; large disagreement between A and B triggers review.
- **Hard safety rules and a safety controller** for extreme readings (automatic stop).
- **Human engineers** make the final stop decision for model-driven alerts, with overrides logged.

- **Abstention:** if critical sensors are missing or invalid, show "insufficient data" and schedule a manual inspection.

Conclusion: Random Forest (B) best balances high failure recall, latency, CPU-only deployment, robustness to missing sensors, explanation needs, external validation and team skills. Highest accuracy (C) is not the best choice because it breaks latency, hardware, explanation and maintenance constraints.

A5: Identifying Hidden Technical Debt (Food Delivery Recommendation System)

> **A5. Prepare a hidden technical debt register and identify the three items that create the greatest production risk.**

Summary register

#	Debt item (category)	Affected component	Owner	Urgency
1	Undeclared consumers	Model score output / 20+ services	ML platform lead + consuming teams	High
2	Unstable data dependencies (renamed event fields)	Event ingestion / feature pipeline	Mobile app team + data engineering	Critical
3	Train-serving mismatch (two distance implementations)	Feature computation	ML engineering	Critical
4	Configuration debt (15 config files, manual production edits)	Ranking weights and thresholds	ML platform / SRE	Critical
5	Glue-code debt (mapping 12 categories to 5)	Output-to-interface adapter	ML engineering + app team	High
6	Pipeline entanglement (undocumented join scripts)	Data joins (customer, restaurant, payment, location, promotion)	Data engineering	High
7	Stale features (hourly availability refresh)	Restaurant availability feature	Data engineering	High
8	Feedback loop (top positions get more clicks)	Training data / ranking	ML team + product	High
9	Selective exposure (clicks treated as preference)	Label definition	ML team + data science	High
10	Monitoring gap (only CTR monitored)	Monitoring	ML platform / MLOps	Critical
11	Ownership gap (no pipeline owner)	Whole pipeline	Engineering leadership	Critical
12	Interface rigidity (5 fixed categories)	Recommendation UI	Product + app team	Medium
13	Hidden post-processing (promotion teams change scores)	Post-inference scoring	Promotions + ML team + leadership	High
14	Duplicated feature definitions	Train/test/production pipelines	ML engineering	High

Detailed register entries

1. Undeclared consumers

- **Evidence:** more than twenty internal services consume model scores, with no central list.
- **Category / component:** undeclared consumers (data dependency) / model output.

- **Failure mode:** a model update changes score scale or meaning and silently breaks downstream services (pricing, marketing, ETA) that assumed the old behaviour; the ML team cannot safely change the model.
- **Consequence:** wrong promotions, broken features, revenue loss; hidden coupling blocks improvement.
- **Detection:** access logs and API-key audit of who calls the score endpoint; dependency tracing.
- **Correction:** a registry of consumers, access control on the score API, versioned score contracts, notice before changes.
- **Owner / urgency:** ML platform lead with consuming teams / High.

2. Unstable data dependencies

- **Evidence:** the mobile team renamed event fields without informing the ML team.
- **Category / component:** unstable data dependency / event ingestion.
- **Failure mode:** features become null or default silently; the model ranks on corrupted inputs (like the notes' UTC-to-local timestamp change).
- **Consequence:** poor recommendations, lower orders; no error is thrown (silent failure).
- **Detection:** schema validation and missingness checks on incoming events; alerts on sudden feature distribution changes.
- **Correction:** data contracts between app and ML teams; schema registry; pipeline pauses on schema violations.
- **Owner / urgency:** mobile app team and data engineering / Critical.

3. Train-serving mismatch

- **Evidence:** training uses one implementation of restaurant distance, production another.
- **Category / component:** train-serving mismatch (skew) / feature computation.
- **Failure mode:** the model sees different distance values in production than in training; learned relationships no longer apply.
- **Consequence:** far restaurants recommended, longer deliveries, cold food, cancellations.
- **Detection:** compare training vs serving feature values for the same requests; skew monitoring.
- **Correction:** a single shared, versioned transformation (feature store) for training and serving; automated consistency tests.
- **Owner / urgency:** ML engineering / Critical.

4. Configuration debt

- **Evidence:** fifteen separate configuration files with ranking weights and thresholds; some edited manually on production servers.
- **Category / component:** configuration debt / ranking parameters.
- **Failure mode:** nobody knows which configuration produced current behaviour; changes are untracked; rollback impossible.
- **Consequence:** unexplained ranking changes, outages, inability to reproduce or audit.
- **Detection:** configuration diff audits; file-integrity monitoring on production servers.
- **Correction:** consolidate into one version-controlled configuration with review, testing and deployment through the pipeline; ban manual production edits.
- **Owner / urgency:** ML platform / SRE / Critical.

5. Glue-code debt

- **Evidence:** a large block of custom glue code maps the new model's 12 categories and continuous score into the old interface's 5 categories.
- **Category / component:** glue code / output adapter.

- **Failure mode:** brittle mapping silently fails when outputs change (like the notes' collections example where everyone defaulted to "Standard").
- **Consequence:** the new model's value is lost or distorted without any error.
- **Detection:** monitor the distribution of final displayed categories vs model outputs.
- **Correction:** redesign the interface to accept the new outputs; replace glue code with a tested, documented adapter; unit tests.
- **Owner / urgency:** ML engineering + app team / High.

6. Pipeline entanglement

- **Evidence:** customer, restaurant, payment, location and promotion data are joined through several undocumented scripts.
- **Category / component:** pipeline jungles / entanglement / data joins.
- **Failure mode:** a change in one source or script alters features elsewhere (changing anything changes everything); failed or delayed joins produce wrong features.
- **Consequence:** silent degradation; privacy risk from undocumented payment/location joins.
- **Detection:** lineage tracking; data-quality tests on join outputs (row counts, key matches).
- **Correction:** document and consolidate joins into an orchestrated, tested pipeline with lineage.
- **Owner / urgency:** data engineering / High.

7. Stale features

- **Evidence:** restaurant availability refreshed hourly although restaurants open and close within minutes.
- **Category / component:** stale features / availability feature.
- **Failure mode:** closed restaurants recommended; open ones hidden.
- **Consequence:** failed orders, poor customer experience, unfair loss of orders for restaurants.
- **Detection:** measure order failures due to closure; feature freshness monitoring.
- **Correction:** streaming or event-driven availability updates; availability filter at serving time.
- **Owner / urgency:** data engineering / High.

8. Feedback loop

- **Evidence:** restaurants shown near the top receive more clicks and therefore appear more often in future training data.
- **Category / component:** direct feedback loop / training data and ranking.
- **Failure mode:** past exposure → observed interaction → training data → future exposure; popular restaurants become more popular regardless of quality.
- **Consequence:** new and small restaurants starved of exposure (fairness to restaurants), less diversity, lock-in.
- **Detection:** track concentration of exposure and new-restaurant exposure over time.
- **Correction:** counterfactual logging of what was shown and at which position; position-bias correction; exploration traffic.
- **Owner / urgency:** ML team + product / High.

9. Selective exposure (clicks as preference)

- **Evidence:** user clicks are treated as direct measures of preference.
- **Category / component:** label definition (proxy label) / training labels.
- **Failure mode:** clicks depend on position, interface and availability, not only preference; the model learns its own ranking.

- **Consequence:** optimizing clicks rather than long-term benefit (satisfaction, repeat orders).
- **Detection:** compare clicks with orders, ratings and retention; A/B tests.
- **Correction:** better labels (completed orders, ratings, repeat purchases), position-debiasing, long-term metrics.
- **Owner / urgency:** ML team + data science / High.

10. Monitoring gap

- **Evidence:** only recommendation click-through rate is monitored.
- **Category / component:** monitoring debt / whole system.
- **Failure mode:** input corruption, skew, stale features, drift and fairness problems are invisible while CTR looks fine.
- **Consequence:** silent failures and revenue loss discovered late.
- **Detection:** (gap itself) audit of monitoring coverage.
- **Correction:** multi-dimensional monitoring: input quality, drift, feature pipelines, output distributions, order conversion, delivery time, cancellations, restaurant exposure fairness, latency, overrides.
- **Owner / urgency:** ML platform / MLOps / Critical.

11. Ownership gap

- **Evidence:** no single person owns the complete pipeline.
- **Category / component:** governance / ownership debt / entire system.
- **Failure mode:** problems fall between teams; no one can approve, stop or roll back.
- **Consequence:** slow incident response; unmanaged risk (NIST Govern failure).
- **Detection:** RACI review; incident post-mortems showing unclear responsibility.
- **Correction:** name an accountable pipeline owner with authority; define team responsibilities.
- **Owner / urgency:** engineering leadership / Critical.

12. Interface rigidity

- **Evidence:** the interface accepts only five fixed categories while the model produces twelve plus a continuous score.
- **Category / component:** rigid UI categories / recommendation interface.
- **Failure mode:** information lost; forces glue code.
- **Consequence:** the new model cannot deliver its value.
- **Detection:** compare model output richness with displayed categories.
- **Correction:** redesign the interface to support new categories and scores.
- **Owner / urgency:** product + app team / Medium.

13. Hidden post-processing

- **Evidence:** promotion teams modify ranking scores after model inference.
- **Category / component:** hidden post-processing / scoring layer.
- **Failure mode:** the displayed ranking is not the model's ranking; evaluation and monitoring measure the wrong thing; the changed rankings feed back into training.
- **Consequence:** unexplained behaviour, possible unfair promotion, misleading metrics.
- **Detection:** log pre- and post-adjustment scores and compare.
- **Correction:** make promotion adjustments explicit, versioned and logged within the pipeline, with governance approval; train on logged exposure.
- **Owner / urgency:** promotions team + ML team + leadership / High.

14. Duplicated feature definitions

- **Evidence:** feature definitions duplicated across training, testing and production pipelines.
- **Category / component:** duplicated logic (skew risk) / feature pipelines.
- **Failure mode:** definitions drift apart, causing train-serving and test-serving mismatch.
- **Consequence:** evaluation results do not reflect production.
- **Detection:** automated comparison of feature outputs across pipelines.
- **Correction:** one centralized, version-controlled feature store (Feast or Tecton, as in the notes).
- **Owner / urgency:** ML engineering / High.

Three items with the greatest production risk:

1. **Train-serving mismatch plus duplicated feature definitions (items 3 and 14):** the production model is effectively running on different inputs from the ones it was trained and tested on, so every offline result is unreliable.
2. **Configuration debt with manual production edits (item 4):** untracked changes make the system irreproducible and impossible to roll back; one mistaken edit can change rankings for all customers.
3. **Monitoring gap combined with the ownership gap (items 10 and 11):** with only CTR monitored and nobody owning the pipeline, all other debts (renamed fields, stale features, broken glue code) can fail silently with no one responsible for detecting or fixing them.

A6: Completing a Problem-Framing Worksheet (University Tutoring Allocation)

> **A6. Complete the problem-framing worksheet for tutoring allocation and answer the follow-up questions.**

Completed statement: At [the end of week 3 of the semester, before tutoring begins], for [every first-year student enrolled on the census date (one student is the unit of analysis)], using only [information recorded by the end of week 3: admission scores, previous academic records, programme of study, course enrolment, attendance in weeks 1 to 3, weekly quizzes 1 to 3, assignment submissions due by week 3, learning-platform activity in weeks 1 to 3, and financial-aid information as recorded at enrolment], produce [a calibrated probability of serious academic difficulty, a ranked list, and an uncertainty flag for borderline or incomplete records] concerning [the remainder of the semester, from week 3 to the release of final results], so that [the academic-support office, with faculty advisers] can [offer the 200 intensive-tutoring places to the highest-ranked students after human review, with lighter support (study-skills workshops, peer mentoring) for the next band]. The system will be compared with [the current process of voluntary requests and faculty referrals] and evaluated using [recall@200, precision@200, calibration and AUROC, reported overall and by subgroup], while accounting for [the higher cost of missing a struggling student, stigma from being flagged, fairness across financial-aid, programme, gender and first-generation groups, the hard limit of 200 places, and governance requirements including student notice, opt-out, appeal, human review and monitoring].

- **Why week 3 is operationally appropriate:** early enough that most of the semester remains for tutoring to change the outcome (the horizon must match the window to act), yet late enough to have real behavioural signals (attendance, first quizzes and submissions). After the midterm the model would be more accurate but "accurate but operationally unhelpful": students may already have failed half the assessments. A defensible variant: allocate about 150 places at week 3 and hold 50 for a week-6 re-scoring of students who start to struggle.
- **Variables available at week 3:** admission scores, previous academic records, programme, course enrolment, attendance (weeks 1 to 3), weekly quizzes 1 to 3, assignment submissions to date, learning-platform activity to date, financial-aid status at enrolment.

- **Variables that cause temporal leakage:** midterm grades (after week 3 unless the index time is post-midterm), final grades (the outcome itself), withdrawal records (part of the outcome), attendance, quizzes, submissions and platform activity after week 3, and any financial-aid change made later (for example aid lost due to poor grades).
- **Defining serious academic difficulty:** Y = 1 if, by the end of the semester, the student fails one or more core first-year courses, or withdraws after the census date, or ends the term with a GPA below the academic-probation threshold. Students who leave for non-academic reasons (medical leave) are recorded separately, and missing outcomes are not coded as 0.
- **Output type:** a calibrated probability used to produce a ranking (not a hard class), because a fixed capacity of 200 makes this ranking under a budget; probabilities also let the team assign lighter support tiers and show unmet need. Recommendations (tutoring vs workshop) follow from risk tiers.
- **Effect of the 200-student limit:** the decision rule is top-200 by $\hat{p}$ after review (k = 200), so the effective threshold is the $\hat{p}$ of the 200th student; add a minimum risk floor so places are not wasted on low-risk students; send students near the cut-off or with incomplete records for human review.
- **False positives:** a place goes to a student who would have passed, displacing someone who needed it; possible stigma and wasted tutor time. **False negatives:** a struggling student gets no support and fails or drops out; this is more costly and harder to reverse, so recall@200 is the primary metric.
- **Groups needing separate analysis:** financial-aid recipients vs non-recipients; programme of study; gender; first-generation students; international and non-native speakers; students with disabilities; commuting or working students; admission route; and intersections. Report selection rate, recall and calibration with sample sizes and confidence intervals.
- **Informing students:** tell all students at orientation that early-warning analytics are used, what data are used, and that the goal is support, not punishment; invitations should be framed positively ("you have been offered free tutoring") and kept confidential.
- **Decline, appeal, human review:** yes. Students may decline tutoring without penalty; students not selected can request support (voluntary route remains open); any student can ask for human review of their data; faculty advisers make the final allocation and can override with a logged reason.
- **Comparison with voluntary requests and faculty referrals:** compare on the same historical cohorts (temporal validation) which approach captures more students who later had serious difficulty (recall@200), and check whether the model finds students who do not seek help because of stigma or awareness. Ideally run a pilot comparing outcomes of students selected by each route, and randomize among borderline students to estimate the causal effect of tutoring.
- **Evidence to justify deployment:** better recall@200 than referrals on temporal validation; good calibration; no subgroup with much worse recall; a prospective silent run; evidence from a pilot that tutoring improves outcomes for flagged students; a model card, notice and appeal process in place; monitoring plan approved.

A7: Identifying Possible Failures in an AI Project (Automated Recruitment Screening)

> **A7. Conduct an end-to-end failure analysis and recommend proceed, redesign, replace or cancel.**

Failure (pipeline stage)	Why it is a failure	Harm	Detection	Prevention / mitigation
Problem definition (formulation)	"Find the best applicants" is undefined: best at what job, measured how, over what horizon? No decision-maker or action defined.	Model optimizes an arbitrary proxy; no way to judge success	Framing review; cannot complete the worksheet	Define the decision (who is interviewed), job-relevant criteria and outcome

Failure (pipeline stage)	Why it is a failure	Harm	Detection	Prevention / mitigation
Target and label selection (formulation/data)	Historical hiring decisions record past recruiter preferences, not quality (status quo bias).	Reproduces past bias at scale	Audit labels vs job performance; adverse-impact analysis	Use a validated job-relevant outcome, or narrow to verified requirements
Data provenance (data)	Unknown how past decisions were made, by whom, under which policies.	Hidden bias and policy changes baked in	Provenance review, datasheet	Document sources; datasheet; exclude unreliable periods
Historical bias (data)	Most hires from five universities; past discrimination in labels.	Candidates from other universities excluded	Selection rates by university and group	Rebalance, remove biased labels, human review
Proxy variables (representation)	Postal code, name, university and employment gaps encode socioeconomic status, ethnicity, gender (gaps for caregiving) and disability.	Indirect discrimination; legal liability	Proxy tests: predict protected attributes from features; ablations	Remove or justify each feature with a job analysis; fairness testing
Population representativeness (data)	Non-traditional applicants under-represented.	Model is unreliable and unfair for them	Subgroup sample sizes and performance	Collect more data; abstain for out-of-distribution applicants
Temporal leakage (representation)	Résumé information submitted after the interview is in the training records.	Inflated accuracy; fails at screening time	Check timestamps and availability time	Snapshot features at application time only
Train-test splitting (evaluation)	Random split within the same year: same roles, recruiters and possibly duplicate applicants on both sides (group and temporal leakage).	Over-optimistic results	Compare with temporal split	Split by time (train on earlier years, test on later) and by applicant
Evaluation metrics (evaluation)	Accuracy alone (96%) with imbalance (few hires) can be achieved by rejecting almost everyone.	Misleading success claims	Report precision, recall, AUROC, selection rates	Pre-registered metrics incl. subgroup and calibration
External and temporal validation (evaluation)	Not tested on another country or later year.	Fails when deployed elsewhere or later	External and temporal holdouts	Validate before any use; restrict scope
Calibration and uncertainty (evaluation/decision)	No uncertainty shown; scores may be overconfident.	Small, meaningless score gaps decide careers	Calibration curves; intervals	Show confidence; abstain on low confidence
Model interpretability (model)	No explanations of rankings.	Recruiters cannot spot proxies; applicants cannot contest	Explanation audits	Interpretable model or stable explanations
Interface design (deployment)	Top ten highlighted in green; ranked list only.	Nudges acceptance; others invisible	Invitation rates by colour/rank	Show full pool, remove colour ranking, show uncertainty
Automation bias (human workflow)	Recruiters trust the list without checking.	Errors pass unchecked	Override rates (very low = rubber-stamping)	Training, required independent review, logged reasons
Human workflow (deployment)	Recruiters review only highlighted applicants.	Human oversight is nominal	Workflow audits	Mandatory review of a wider pool; staffing
Appeals and procedural fairness (governance)	Applicants cannot get an explanation or appeal.	Unchallengeable, possibly unlawful decisions	Complaints, legal review	Notice, explanation, appeal and human re-review

Failure (pipeline stage)	Why it is a failure	Harm	Detection	Prevention / mitigation
Feedback loops (monitoring)	Model-selected hires become new training data.	Bias amplified over time	Track selection concentration over retrains	Counterfactual logging; do not retrain on own selections
Selective labels (data/monitoring)	Outcomes only observed for those hired.	Rejected groups never get positive labels	Compare exposure vs outcomes	Random audits of rejected applicants; better labels
Retraining (lifecycle)	Automatic six-monthly retraining with no validation gate.	Worse or more biased versions deployed	Version comparison	Model validation gate; human approval; rollback
Monitoring (monitoring)	No monitoring of subgroups, drift, overrides or loops.	Silent failures and discrimination	(Absence itself)	Multi-dimensional monitoring dashboard
Accountability and governance (governance)	No person or committee responsible.	No one can stop harm (NIST Govern failure)	Governance review	Named accountable owner, committee, audits
Legal, ethical, organizational constraints	Possible violation of anti-discrimination and data-protection law; reputational risk.	Lawsuits, fines, loss of trust	Legal and ethics review	Compliance review before any pilot; model card; impact assessment

Recommendation: Redesign the project before any pilot.

A controlled pilot is not justified now, because the core problems are in the foundations: the objective is undefined, the label encodes historical bias, proxies and post-interview leakage contaminate the features, the evaluation (random split, accuracy only, no baseline, no external or temporal validation) cannot support any claim, and there is no governance or appeal. Piloting would expose real applicants to discrimination and generate more biased data through the feedback loop. Cancelling outright is not necessary if a legitimate need exists (screening a large volume of applications), and a simpler process may be part of the redesign (for example, structured rule-based checks of verified requirements plus human review). The redesigned project must: define the decision and a job-relevant outcome; remove proxies and leaked features; use temporal and external validation with subgroup, calibration and baseline comparison against recruiter review; redesign the interface (full pool, uncertainty, no green highlighting); add notice, explanation and appeal; stop automatic retraining; add monitoring; and assign a named accountable owner. Only then should it proceed to a silent, then advisory, controlled pilot.

Part 9: Last-Minute Revision

9.1 What is most likely to come

Based on how the notes and the practice set are built, these are the highest-probability quiz items:

- **Produce or critique an artifact:** a problem-framing worksheet, a Model Card (heart-disease template), a Datasheet, or a failure analysis. Know the section headings by heart.
- **Spot the flaw in a scenario:** temporal leakage, train-serving mismatch, proxy encoding, feedback loop and selective labels, aggregation bias, random split for a temporal claim, accuracy-only evaluation, automation bias from interface design.
- **Choose a model or threshold:** explain why the most accurate model is not always the best (loan and predictive-maintenance examples), and how C(t), capacity and asymmetric costs set t.
- **Definitions and comparisons:** symbolic vs statistical, prediction vs causal reasoning, discrimination vs calibration, autonomy vs intelligence, AI vs ML vs DL vs data science, generative vs discriminative, the four Russell-Norvig views, the four autonomy levels, NIST Govern-Map-Measure-Manage.
- **Case studies:** Google Flu Trends, Epic Sepsis Model, TFX, Obermeyer (cost as proxy), predictive policing.
- **One question from your assignment or notebook:** RYDA index time and leakage, why ranking with k = 300, WePrep subgroup gaps and recommendation, datasheet suitability, résumé feedback loop and neutralization debt, MLP forward pass by hand, Bayes posterior, O(N) vs O(1).

9.2 One-line answers to memorize

If asked...	Say...
Where must an AI project start?	With the decision or capability, not the algorithm or dataset.
What makes a feature legal to use?	It is recorded and available at or before the index time.
Why not a random split?	For temporal claims it leaks the future; use temporal validation (train past, test future).
Why not accuracy alone?	It ignores asymmetric costs, calibration, subgroups, imbalance, baselines and decision utility.
Discrimination vs calibration?	Ranking positives above negatives vs predicted probabilities matching observed frequencies.
Why probabilities over hard labels?	They keep uncertainty and let the threshold follow costs and capacity.
How to set t?	Minimize C(t) = c_FP·P(FP|t) + c_FN·P(FN|t) + c_delay·E[D|t] + c_workflow(t), within capacity, fairness and human review.
Capacity-limited allocation?	Rank and take the top k; evaluate with precision@k and recall@k.
Train-serving mismatch?	Different preprocessing in training vs production (median vs 0 imputation).
Proxy encoding?	Neutral features (postal code, university) encode protected attributes.
Feedback loop in ranking?	Past exposure → observed interaction → training data → future exposure.
Selective labels?	Outcomes only observed for cases the system (or people) chose to look at.
Why optimization is blind?	It solves the specified objective; it cannot validate the target, sample, loss or ethics.
Autonomy vs intelligence?	Thermostat: autonomous, not intelligent. LLM: intelligent, no agency.

If asked...	Say...
Socio-technical layers?	Model, Data Infrastructure, Interface, Human Workflow, Institutional Rules.
NIST functions?	Govern (accountability), Map (identify risks), Measure (assess), Manage (act).
Model Card sections?	Overview; Data Description; Performance Metrics (incl. subgroups, fairness); Known Limitations; Intended and Inappropriate Uses; Ethical Considerations; Additional Considerations (future work).
Datasheet sections?	Motivation; composition; collection process; preprocessing; recommended uses; limitations (plus maintenance).
8 pipeline stages?	Problem, Data, Representation, Model, Evaluation, Decision, Deployment, Monitoring (iterative).
8 failure modes?	Wrong problem, poor data, leakage, weak evaluation, underspecification, distribution shift, ignored workflow, no monitoring.
Leakage types?	Target, temporal, group, preprocessing, label, benchmark contamination.
Occam's razor?	A heuristic: avoid unnecessary complexity, not necessary complexity (TB + arthritis).
Google Flu Trends lesson?	Volume does not remove measurement bias; proxies drift; anchor to ground truth; keep lineage.
Epic Sepsis lesson?	External validation; label must match the event; timing matters; alert volume is a system metric.
TFX lesson?	A training script is not a production system; schemas, validation, shared transforms, lineage, model validation.

9.3 Common traps

- Writing "the AI decided" instead of naming the human and institutional choices behind it.
- Using a default threshold of 0.5 without justification.
- Saying "high accuracy, so deploy" without calibration, subgroups, baseline and workflow.
- Forgetting exclusions, the unit of analysis, and the horizon in a problem statement.
- Treating missing or unobserved outcomes as negatives.
- Assuming high risk means high benefit from an intervention (a causal question).
- Forgetting the recommendation line at the end of a Model Card or failure analysis.

9.4 Numbers worth remembering from the notes

- Epic Sepsis: AUROC 0.63; 18% of hospitalizations alerted; 67% of sepsis cases missed at score ≥ 6; 183 cases found before timely antibiotics.
- Google Flu Trends: overestimated in 100 of 108 weeks; more than 2× CDC in February 2013; about 50 million terms vs 1,152 observations; combined model MAE 0.232 vs 0.486 (GFT) and 0.311 (lagged CDC).
- Loan example: DNN 94% vs logistic regression 91% with 15 features; about 50,000 applications per year.
- Baggage screening: t = 0.4 gives 96% accuracy but breaks the checkpoint; t = 0.7 gives 94% and minimizes C(t).
- Readmission example: random split 85% vs temporal 68%; AUC 0.82 overall vs 0.55 for non-English speakers; 80% predicted vs 35% observed.
- Notes' Model Card example: accuracy 85%, precision 0.80, recall 0.75, F1 0.77; male recall 0.78 vs female 0.70.

- A2 scenario: accuracy 86%, precision 0.82, recall 0.74, F1 0.78; male recall 0.80, female 0.62, under-40 0.55.

Good luck with the quiz.